# Survey Question Harmonization Constraints

AI-Assisted Analysis of Federal Survey Consolidation Potential

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2026-02-10

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# Chapter 1

# Survey Question Harmonization Constraints

AI-Assisted Analysis of Federal Survey Consolidation Potential

DRAFT Notice

**DRAFT** — These materials are in development and have not been independently validated. The author makes no claims regarding validity until formal review is complete.

Disclaimer

The views expressed in this report are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce. This is exploratory research conducted to assess the feasibility of AI-assisted survey analysis methods.

# Preface

This report presents findings from an AI-assisted analysis of federal survey question consolidation potential. Using a multi-model ensemble approach, we analyzed 1,598 question pairs from the Current Population Survey (CPS) and FoodAPS against the American Community Survey (ACS) to determine:

1. **What proportion of survey questions can be consolidated?**
2. **What barriers prevent consolidation?**
3. **Can AI-assisted methods accelerate this traditionally labor-intensive analysis?**

## 1.1 Key Findings

- ~**44% consolidation potential** identified across 380 questions
- **97% of failures** stem from construct/concept differences (not fixable)
- **AI handles 76%** of classifications with high confidence, routing only 24% to expert review
- **Reproducible methodology** applicable to other survey harmonization efforts

## 1.2 Report Structure

- **Chapters 1-3**: Background, research context, and harmonization framework
- **Chapters 4-6**: Methodology, results, and discussion
- **Appendices**: Technical details, taxonomy definitions, and expert review tables

## 1.3 Data Availability

All analysis code, intermediate outputs, and expert review tables are available in the project repository: `reports/03_harmonization_constraints/`

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**Note:** This is Report 03 of a multi-phase project analyzing federal survey questions. Reports 01 (concept classification) and 02 (pairwise comparison setup) established the foundation for this harmonization analysis.

# Executive Summary

## Problem

The federal survey ecosystem collects overlapping data across agencies that currently exists in silos. These overlaps represent untapped analytical potential: harmonizable questions serve as **bridge variables** enabling cross-survey data integration, which increases explanatory power without additional data collection. As survey response rates decline, extracting more value from existing data becomes critical. Determining which questions are harmonizable requires expert judgment and traditionally takes weeks or months of manual review. A secondary benefit: overlaps also represent opportunities for survey consolidation to reduce respondent burden.

## Approach

We developed an AI-assisted pairwise analysis system using: - **Multi-model ensemble** (OpenAI, Anthropic, Google) for initial classification - **Arbitration pattern** to resolve disagreements - **Question-level rollup** using two-axis triage (Borda direction × Entropy stability) - **1,598 question pairs** analyzed (CPS-ACS, FoodAPS-ACS)

## Key Results

- **CPS**: 41.7% harmonizable (100 of 240 questions)
- **FoodAPS**: 48.6% harmonizable (68 of 140 questions)
- **Combined**: ~44% harmonization rate

These **168 linkage-ready questions** represent potential bridge variables for cross-survey data enrichment. Harmonization feasibility levels characterize bridge variable quality: - **F1** (direct harmonization): High-quality bridge variables usable for statistical matching without adjustment - **F2** (harmonization with transformation): Usable bridge variables requiring methodological adjustment - **F3** (not harmonizable): Characterized linkage constraints defining where cross-survey enrichment is not viable

**Barrier Analysis:** - 97% of F3 pairs = Construct/Concept (CC) differences - CC.1 (concept definition) accounts for 70% of all incompatible pairs - These barriers validate survey specialization: most non-harmonizable pairs serve fundamentally different analytical purposes, defining precisely where linkage works and where it doesn't

**Expert Review Load:** - 76% auto-processed with high confidence - 24% (93 questions) routed to expert review - AI handles breadth; humans judge edge cases

## Deliverables

1. **168 harmonizable question pairs** characterized as potential bridge variables for cross-survey enrichment
2. **212 non-harmonizable pairs** with linkage quality constraints documented via barrier codes
3. **93 questions flagged for expert review** (uncertain or contested cases)
4. **Reproducible methodology** applicable to entire federal survey ecosystem

## Implications

This work demonstrates that AI-assisted methods can: - **Identify** bridge variables for cross-survey data integration, increasing analytical power of existing data - **Characterize** linkage quality constraints, defining where enrichment is viable - **Accelerate** harmonization analysis (days instead of weeks) - **Preserve** human oversight and expert judgment - **Scale** to analyze 7,000+ questions across the federal survey ecosystem

For agencies considering survey consolidation, the same analysis identifies where instrument streamlining is technically feasible as a burden reduction strategy.

## Next Steps

1. Expert validation of bridge variable quality ratings
2. Report 04: AI-assisted discovery of cross-survey enrichment patterns using bridge variable catalog
3. Enrichment pilots: test statistical matching using identified bridge variables
4. Expansion to additional federal surveys
5. Application to survey editing and imputation workflows

# Chapter 2

# Introduction

## 2.1 Problem Statement

The federal statistical system conducts dozens of surveys annually, many asking overlapping questions about demographics, economics, health, and social conditions. These overlaps represent both analytical opportunities and challenges:

1. **Untapped analytical potential**: Overlapping questions across surveys represent bridge variables that could enable cross-survey statistical linkage, increasing explanatory power without additional data collection. Currently, this potential remains largely unrealized due to the difficulty of systematically identifying harmonizable questions across the ecosystem.
2. **Declining response rates**: As survey participation drops (ACS response rates declined from mid-90s to ~85%, CPS around 70%), extracting more value from existing data becomes critical. Cross-survey enrichment through linkage provides this — using bridge variables to integrate datasets and expand analytical capabilities.
3. **Respondent burden**: Overlaps also represent potential for survey streamlining. Same households answering similar questions across multiple surveys creates unnecessary burden, and consolidation where feasible offers a secondary benefit.

Determining which questions are harmonizable — and characterizing the quality of potential bridge variables — requires deep subject-matter expertise and traditionally involves weeks or months of manual review by survey methodologists.

## 2.2 Research Questions

This report addresses three core questions:

### 2.2.1 RQ1: Harmonization Potential

**What proportion of federal survey questions are harmonizable — serving as potential bridge variables for cross-survey data enrichment?**

- Hypothesis: Significant harmonization potential exists but requires systematic identification
- Approach: Exhaustive pairwise comparison of CPS/FoodAPS questions against ACS
- Outcome: Quantify F1 (direct harmonization), F2 (harmonization with adjustment), F3 (not harmonizable) rates and identify linkage-ready bridge variables

### 2.2.2 RQ2: Linkage Quality Constraints

**What barriers prevent survey question harmonization, and how do these constraints define where cross-survey linkage is viable?**

- Hypothesis: Most barriers stem from construct differences, not operational issues; these barriers characterize linkage quality constraints
- Approach: Classify incompatibilities using DataSHaPER/Maelstrom harmonization framework
- Outcome: Identify primary barrier types, their prevalence, and implications for bridge variable quality

### 2.2.3 RQ3: AI-Assisted Methods

**Can AI-assisted methods accelerate this traditionally labor-intensive analysis?**

- Hypothesis: LLM ensemble can handle routine classifications, reserving expert judgment for edge cases
- Approach: Multi-model rating + arbitration + two-axis triage
- Outcome: Measure agreement rates, expert review load reduction

## 2.3 Scope

### 2.3.1 Surveys Analyzed

- **American Community Survey (ACS)**: Target survey with comprehensive demographic/economic questions
- **Current Population Survey (CPS)**: Employment-focused survey (240 questions)
- **FoodAPS**: Food security survey (140 questions)

Total: **380 unique source questions** compared against ACS, yielding **1,598 question pairs**.

### 2.3.2 Framework Applied

We adopt the DataSHaPER/Maelstrom retrospective harmonization framework: - **F1**: Direct recode (mechanically transformable) - **F2**: Statistical adjustment (requires modeling/bridging) - **F3**: Incompatible (fundamental barriers prevent harmonization)

### 2.3.3 What This Report Is NOT

- Not a recommendation to eliminate any surveys (policy decision)

- Not a claim that all harmonizable pairs should be linked — fitness-for-purpose assessment is required for each enrichment use case
- Not a replacement for expert judgment (AI assists, experts decide)

## 2.4 Prior Work

This is **Report 03** of a multi-phase project:

- **Report 01**: Concept classification of 7,400+ questions using LLM ensemble
- **Report 02**: Pairwise comparison setup and sampling strategy
- **Report 03** (this report): Harmonization constraints and consolidation analysis

Each report builds on the previous, with validated outputs feeding forward.

## 2.5 Report Structure

The remainder of this report is organized as:

- **Chapter 2 (Background)**: Survey harmonization frameworks and prior work
- **Chapter 3 (Methodology)**: 5-stage pipeline from rating to deliverables
- **Chapter 4 (Results)**: Consolidation rates, barrier analysis, agreement statistics
- **Chapter 5 (Discussion)**: Interpretation, limitations, implications
- **Chapter 6 (Conclusion)**: Summary and future directions
- **Appendices**: Taxonomy definitions, methodology decisions, expert review tables

## 2.6 Key Contributions

This work contributes:

1. **Bridge variable catalog** identifying harmonization potential across federal surveys (~44% harmonizable), characterizing 168 linkage-ready questions for cross-survey data enrichment
2. **Linkage quality characterization** showing CC dominance (97% of incompatibilities), defining precisely where cross-survey enrichment is viable and where surveys serve distinct analytical purposes
3. **Operational triage framework** (two-axis: Borda direction × Entropy stability) enabling efficient expert review prioritization
4. **Reproducible methodology** for AI-assisted survey harmonization scalable to the entire federal survey ecosystem

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**Note**: All data, code, and intermediate outputs are available in `reports/03_harmonization_constraints/` for replication and extension.

# Chapter 3

# Background

## 3.1 Survey Harmonization Frameworks

### 3.1.1 DataSHaPER/Maelstrom Framework

The DataSHaPER (Data Schema and Harmonization Platform for Epidemiological Research) framework, developed by Fortier et al. (2011) and Fortier et al. (2017), provides a structured approach to retrospective data harmonization. This framework classifies variable pairs into three feasibility categories:

| Code | Feasibility | Definition | Action Required |
|---|---|---|---|
| **F1** | **Direct recode** | Variables are mechanically transformable through simple recoding, collapsing categories, or unit conversion | Simple data transformation |
| **F2** | **Statistical adjustment** | Variables require modeling, imputation, or bridging studies to make comparable | Statistical harmonization methods |
| **F3** | **Incompatible** | Variables measure fundamentally different constructs and cannot be harmonized without re-fielding | No harmonization possible |

Fortier et al. (2011) applied this framework across 53 epidemiological studies and found that 38% of variables could be harmonized through direct or statistical methods.

### 3.1.2 Barrier Taxonomy

When variables are classified as F3 (incompatible), the framework identifies specific barrier types:

#### 3.1.2.1 Construct/Concept (CC)

- **CC.1**: Concept definition differences (e.g., "employment" including/excluding unpaid work)
- **CC.2**: Operationalization differences (different behavioral indicators for same concept)
- **CC.3**: Boundary conditions (different thresholds or cutoffs)
- **CC.4**: Scope inclusions (different components counted)

#### 3.1.2.2 Temporal Constraints (TC)

- **TC.1**: Reference period length (7-day vs 12-month recall)
- **TC.2**: Temporal framing (point-in-time vs habitual vs retrospective)
- **TC.3**: Calendar alignment (fixed vs rolling reference periods)

#### 3.1.2.3 Response Scale (RS)

- **RS.1**: Scale type (binary vs Likert vs continuous)
- **RS.2**: Category structure (different number/boundaries)
- **RS.3**: Anchoring/labels (different verbal anchors)
- **RS.4**: Numeric vs verbal scales

#### 3.1.2.4 Population/Coverage (PC)

- **PC.1**: Universe definition (target population differs)
- **PC.2**: Frame exclusions (different sampling exclusions)
- **PC.3**: Age bounds (different age eligibility)
- **PC.4**: Geographic scope (different coverage)

#### 3.1.2.5 Mode/Context (MC)

- **MC.1**: Interview mode (CATI vs web vs in-person)
- **MC.2**: Question routing (different skip patterns)
- **MC.3**: Contextual priming (question order effects)
- **MC.4**: Proxy response rules

#### 3.1.2.6 Processing/Metadata (PM)

- **PM.1**: Coding schemes (different classification systems)
- **PM.2**: Derived variables (different construction algorithms)
- **PM.3**: Documentation gaps (insufficient metadata)

**See Appendix A** for complete taxonomy definitions with examples.

## 3.2 Cross-Survey Data Enrichment

Harmonized survey questions serve a dual purpose beyond instrument consolidation: they function as **bridge variables** enabling cross-survey statistical linkage and data enrichment.

**Statistical data fusion** (also called synthetic matching or data integration) is an established technique that combines information from multiple data sources when direct record linkage is not possible (D'Orazio et al., 2006; Rässler, 2002). The key requirement is a set of overlapping variables — bridge variables — measured in both datasets that enable statistical matching.

In the federal survey ecosystem, overlapping questions across surveys represent potential bridge variables. However, the sheer scale of the system (7,000+ questions across 48 surveys) has prevented systematic identification of these opportunities. Individual researchers identify linkage possibilities through domain knowledge and manual review, but no infrastructure exists for discovering cross-survey enrichment patterns at scale.

As federal survey response rates decline (ACS from mid-90s to ~85%, CPS around 70%), extracting more analytical value from existing data becomes increasingly critical. Cross-survey enrichment through bridge variables offers this capability without requiring additional respondent contact or data collection. The DataSHaPER harmonization framework — originally developed for epidemiological research — provides quality ratings for potential bridge variables (F1 = direct use, F2 = methodological adjustment needed, F3 = not viable).

This analysis demonstrates that AI-assisted harmonization analysis can systematically identify bridge variables across the federal survey ecosystem, enabling data enrichment strategies that complement traditional consolidation approaches.

## 3.3 Related Literature

### 3.3.1 Survey Data Harmonization

- Wolf et al. (2016): Harmonizing survey questions between cultures and over time
- Saris & Gallhofer (2014): Design, evaluation, and analysis of questionnaires for survey research
- Slomczynski & Tomescu-Dubrow (2018): Basic principles of survey data recycling

### 3.3.2 AI-Assisted Survey Analysis

### 3.3.3 Expert Aggregation and Multi-Rater Methods

The methodological approach in this report—independent LLM raters with structured arbitration—draws from established traditions in expert aggregation and inter-rater reliability assessment.

#### 3.3.3.1 The Delphi Method

The Delphi method, developed at RAND Corporation in the 1950s for technology forecasting, formalized a structured approach to expert judgment aggregation (Dalkey & Helmer, 1963). The method's core principles remain relevant:

- **Independent elicitation**: Experts provide judgments without knowledge of others' responses, preventing anchoring bias
- **Structured iteration**: Disagreements are systematically surfaced and resolved through controlled feedback
- **Anonymity**: Removes social pressure that can distort group judgment

Linstone & Turoff (1975) codified these principles in *The Delphi Method: Techniques and Applications*, which became the canonical reference for applications ranging from policy analysis to technology assessment. The method gained particular traction in the 1970s for problems where expert judgment was necessary but consensus was difficult to achieve through traditional deliberative processes.

#### 3.3.3.2 Ensemble Methods in Machine Learning

The Netflix Prize competition (2006-2009) demonstrated that combining predictions from multiple independently-tuned algorithms consistently outperforms any single model (Bell & Koren, 2007). The winning solution achieved a 10% improvement in recommendation accuracy through ensemble methods rather than novel algorithmic approaches (Lohr, 2009).

This principle—that aggregating diverse estimators reduces prediction variance—underlies random forests (Breiman, 2001), boosting methods, and modern neural network ensembles. The key insight is that models with different architectures, training procedures, or optimization trajectories produce partially decorrelated errors, and averaging across them cancels noise while preserving signal.

#### 3.3.3.3 Application to LLM Classification

Large language models trained by different organizations (Anthropic, OpenAI, Google) exhibit systematic differences in classification behavior due to:

- Different pre-training corpora and curation decisions
- Different alignment and fine-tuning procedures
- Different architectural choices and model scales

These differences create the *error decorrelation* that makes ensemble methods effective. A classification that all three models agree on is more likely to reflect genuine signal than idiosyncratic model behavior.

Our arbitration pipeline operationalizes this insight: independent LLM classifications (analogous to anonymous Delphi expert judgments), followed by structured disagreement resolution (analogous to controlled Delphi iteration). The approach is not methodologically novel—it is a computational implementation of principles established in decision science and ensemble learning over the past seven decades.

## 3.4 Prior Work: Reports 01-02

### 3.4.1 Report 01: Concept Classification

- Mapped 7,400+ survey questions to Census Bureau taxonomy
- Used LLM ensemble (Claude Haiku + GPT-4o-mini) with arbitration
- Achieved 99.5% categorization success rate
- Cohen's = 0.842 (almost perfect agreement)

### 3.4.2 Report 02: Pairwise Comparison Setup

- Identified 1,598 question pairs for harmonization analysis
- Selected CPS and FoodAPS as representative source surveys
- Developed sampling strategy based on concept categories
- Established pair-level comparison methodology

### 3.4.3 Building Forward

Report 03 (this report) takes the validated pairwise comparisons from Report 02 and:

1. Classifies each pair using harmonization framework (F1/F2/F3)
2. Identifies barrier codes for F3 pairs
3. Rolls up pair-level results to question-level consolidability
4. Generates expert review tables for stakeholder use

## 3.5 Census Bureau Context

The Census Bureau conducts multiple surveys covering demographics, economics, housing, and social conditions:

- **ACS**: Comprehensive demographic/economic data, ~3.5M households annually
- **CPS**: Labor force statistics, ~60K households monthly
- **SIPP**: Household dynamics and program participation
- **Others**: American Housing Survey, Survey of Income and Program Participation, etc.

**Declining response rates** create urgency for extracting more value from existing data. ACS response rates have declined from the mid-90s to approximately 85%, while CPS response rates hover around 70%. This trend makes cross-survey data enrichment increasingly valuable: harmonized questions serve as bridge variables enabling statistical linkage, which expands analytical capabilities without requiring additional respondent contact.

Harmonization across these surveys enables:

- **Cross-survey data enrichment**: Using bridge variables to integrate datasets and increase explanatory power
- **Improved inference**: Combining specialized survey content for richer analysis
- **Reduced respondent burden**: Secondary benefit through potential survey consolidation where feasible
- **Improved data quality**: Consistent measurement across surveys
- **Resource efficiency**: Coordinated questionnaire design

## 3.6 Research Gap

While survey harmonization frameworks exist (DataSHaPER, SDR methodology), **no prior work has**:

1. Characterized federal survey overlaps as potential bridge variables for statistical data fusion and cross-survey enrichment
2. Applied harmonization frameworks systematically to federal survey questions at scale

3. Used AI-assisted methods to accelerate the classification process
4. Quantified harmonization potential across major federal surveys and identified linkage-ready bridge variables
5. Developed operational triage frameworks for expert review prioritization

This report addresses these gaps, with primary focus on identifying bridge variables for cross-survey data enrichment and secondary findings on consolidation potential.

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**Key Takeaway**: Survey harmonization is well-established in epidemiology and international survey research, but has not been systematically applied to the U.S. federal survey ecosystem. This report demonstrates that AI-assisted methods can scale this analysis.

# Chapter 4

# Methodology

## 4.1 Overview

We developed a 5-stage pipeline for AI-assisted survey harmonization analysis. The pipeline characterizes harmonization potential across federal surveys, identifying bridge variables for cross-survey data enrichment and characterizing linkage quality constraints that define where cross-survey integration is viable.

```
Stage 1: Rating (3-model ensemble)
   ↓
Stage 2: Agreement Analysis (inter-rater reliability)
   ↓
Stage 3: Arbitration (disagreement resolution)
   ↓
Stage 4: Question-Level Rollup (best-match + triage)
   ↓
Stage 5: Deliverables (expert review tables)
```

All code is available in `reports/03_harmonization_constraints/scripts/`.

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## 4.2 Stage 1: Rating

### 4.2.1 Objective

Classify 1,598 question pairs using harmonization framework (F1/F2/F3 + barrier codes).

### 4.2.2 Multi-Model Ensemble

We employed three frontier LLMs as independent raters:

| Model | Version | Behavior Profile |
|---|---|---|
| **OpenAI** | gpt-4o-mini | Moderate synthesis, slight self-consistency bias |
| **Anthropic** | claude-3-5-haiku-20241022 | High synthesis, neutral |
| **Google** | gemini-2.0-flash-exp | Deferential, conservative |

**Why three models?** - Reduces single-model bias - Different training data and architectures - Behavioral diversity improves coverage

### 4.2.3 Prompt Design

Each model received: 1. **Harmonization framework definitions** (F1/F2/F3) 2. **Barrier taxonomy** (CC, TC, RS, PC, MC, PM) 3. **Question pair** (source + target text) 4. **Task**: Classify feasibility + provide reasoning

**Output format** (structured JSON):

```
{
  "feasibility": "F1|F2|F3",
  "barrier_code": "CC.1|TC.2|..." (if F3),
  "confidence": "HIGH|MODERATE|LOW",
  "reasoning": "Explanation..."
}
```

### 4.2.4 Execution

- **Parallel processing**: 6 workers per model
- **Batch size**: 10 question pairs per API call
- **Error handling**: Exponential backoff for rate limits
- **Checkpointing**: Resume capability for interrupted runs
- **Output**: `output/results/stage1_{model}_results.jsonl`

### 4.2.5 Data Quality

| Metric | OpenAI | Anthropic | Google |
|---|---|---|---|
| **Pairs rated** | 1,598 | 1,598 | 751 (47%) |
| **Success rate** | 100% | 100% | 47% |
| **Avg confidence** | 0.89 | 0.91 | 0.87 |

**Note**: Google rate-limited at 751 pairs. Remaining analysis uses OpenAI-Anthropic two-way comparison with Google as validation subset.

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## 4.3 Stage 2: Agreement Analysis

### 4.3.1 Objective

Quantify inter-rater reliability and identify disagreements for arbitration.

### 4.3.2 Metrics Calculated

#### 4.3.2.1 Cohen's Kappa (Pairwise)

- **OpenAI-Anthropic**: = 0.845 (almost perfect)
- **OpenAI-Google**: = 0.796 (substantial)
- **Anthropic-Google**: = 0.833 (almost perfect)

#### 4.3.2.2 Fleiss' Kappa (Three-Way)

- **All three models** (751 pairs with Google): = 0.833

**Interpretation**: High agreement validates that harmonization judgments are consistent across models, even with different architectures and training data.

### 4.3.3 Agreement Patterns

| Level | OpenAI-Anthropic | Description |
|---|---|---|
| **Topic (L1)** | 91.2% | High-level feasibility |
| **Subtopic (L2)** | 84.5% | Barrier category |
| **Full code** | 78.9% | Complete barrier sub-code |

**Finding**: Models agree more on broad feasibility than specific barrier codes, consistent with hierarchical classification difficulty.

### 4.3.4 Disagreement Identification

- **Pairs needing arbitration**: 339 (21.2% of 1,598)
- **Criteria**: OpenAI Anthropic on feasibility or barrier code
- **Output**: `output/analysis/arbitration_candidates.csv`

---

## 4.4 Stage 3: Arbitration

### 4.4.1 Objective

Resolve disagreements through higher-capability model arbitration.

### 4.4.2 Arbitration Models

| Model | Version | Role |
|---|---|---|
| **OpenAI** | gpt-4o | Arbitrator (stronger than rater) |
| **Anthropic** | claude-3-5-sonnet-20241022 | Arbitrator |
| **Google** | gemini-2.0-flash-thinking-exp | Arbitrator (experimental reasoning) |

### 4.4.3 Arbitration Prompt

Arbitrators received: 1. **Source and target question text** 2. **Initial ratings** from 2-3 raters (with reasoning) 3. **Disagreement summary** 4. **Task**: Provide final verdict with detailed justification

### 4.4.4 Voting Logic

When arbitrators disagree, final verdict determined by: 1. **Majority vote** (if 2+ agree) 2. **Borda count** (ranked preference if no majority) 3. **Entropy check** (flag high disagreement for expert review)

### 4.4.5 Arbitration Results

| Metric | Value |
|---|---|
| **Pairs arbitrated** | 339 |
| **Arbitrator agreement** | 87.6% |
| **Confidence distribution** | HIGH: 68%, MODERATE: 24%, LOW: 8% |

**Quality check** (11 validation tests): All passed

### 4.4.6 Behavioral Findings

- **OpenAI**: Tends toward F2 when uncertain (optimistic about statistical adjustment)
- **Anthropic**: Balanced, detailed reasoning
- **Google**: Conservative, often defers to initial raters

**See**: `docs/FINDINGS_R03_S3_001_arbitration_analysis.md` for detailed behavioral analysis.

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## 4.5 Stage 4: Question-Level Rollup

### 4.5.1 Objective

Convert 1,598 pair-level results to 380 question-level consolidability assessments with triage.

### 4.5.2 Challenge: Pair-Level Inflation

Each of 380 source questions was compared against multiple ACS questions: - **Average**: 4.2 comparisons per question - **Range**: 1-12 comparisons per question - **Problem**: One F1 match makes a question consolidable, even if 11 other comparisons failed

**Solution**: Best-match rollup + two-axis triage.

### 4.5.3 Best-Match Selection

For each source question: 1. Identify all pairs involving that question 2. Select pair with highest consolidability: - F1 > F2 > F3 (feasibility hierarchy) - Within same feasibility, highest Borda score 3. Output: One best match per question

**Result**: 380 question-level records with best ACS match.

### 4.5.4 Scoring Methods (Ensemble Approach)

We computed four complementary scores for each pair:

#### 4.5.4.1 1. Composite Score (Baseline)

```
score = {
  1.0 if unanimous F1/F2,
  0.5 if majority F1/F2,
  0.0 if unanimous F3,
  NA otherwise
}
```

#### 4.5.4.2 2. Entropy Score (Stability)

```
H = -Σ p(x) log  p(x)
normalized to [0, 1]
```

- Low entropy = high agreement (stable)
- High entropy = disagreement (unstable)

#### 4.5.4.3 3. Bayesian Score (Probabilistic)

```
P(consolidable | ratings) = (positives +  ) / (total +   + )
```

- Incorporates prior beliefs
- Shrinks toward population mean

#### 4.5.4.4 4. Borda Count (Direction)

```
Borda = Σ ranks / max_possible_ranks
```

- F1 = 2 points, F2 = 1 point, F3 = 0 points
- Captures direction of ensemble

### 4.5.5 Two-Axis Triage Framework

**Purpose**: Separate “what’s the answer?” (Borda) from “how much did they argue?” (Entropy).

**Thresholds**: Median split on question-level best-match scores - Borda median: 0.167 - Entropy median: 0.330

**Why question-level medians?** - Pair-level distribution dominated by unanimous F3 pairs (median 0) - Question-level reflects actual decision space for triage - N=380 questions, not N=1,598 pairs

**Quadrant Definitions**:

| Quadrant | Borda | Entropy | Interpretation | Action |
|---|---|---|---|---|
| **Q1** | High | High | Confident consolidable | Auto-accept, spot-check |
| **Q2** | Low | High | Confident non-consolidable | Auto-reject, low priority |
| **Q3** | High | Low | Uncertain accept (leaning yes but contested) | **Expert review priority** |
| **Q4** | Low | Low | Uncertain reject (genuinely ambiguous) | Expert review secondary |

**Rationale**: - Q1 (High Borda, High Entropy): Models agree it’s consolidable → trust - Q2 (Low Borda, High Entropy): Models agree it’s not consolidable → trust - Q3 (High Borda, Low Entropy): Leaning yes but unstable → expert validate - Q4 (Low Borda, Low Entropy): Unclear direction → expert clarify

### 4.5.6 Triage Assignment Results

| Quadrant | Count | % | Description |
|---|---|---|---|
| Q1 | 151 | 39.7% | Auto-accept (confident consolidable) |
| Q2 | 136 | 35.8% | Auto-reject (confident non-consolidable) |
| Q3 | 40 | 10.5% | Expert review (uncertain accept) |
| Q4 | 53 | 13.9% | Expert review (uncertain reject) |

**Expert review load**: 24.5% (93 of 380 questions)

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## 4.6 Stage 5: Deliverables

### 4.6.1 Objective

Generate stakeholder-ready outputs for expert review and implementation.

### 4.6.2 Expert Review Tables

Three CSV files generated: 1. `expert_review_cps.csv` (240 CPS questions) 2. `expert_review_foodaps.csv` (140 FoodAPS questions) 3. `expert_review_combined.csv` (380 questions, all surveys)

**Columns included**: - Source question text - Best ACS match text - Feasibility (F1/F2/F3) - Barrier code (if F3) - Borda score (direction) - Entropy score (stability) - Triage quadrant (Q1-Q4) - Confidence flags - Reasoning summary

### 4.6.3 Visualizations

All figures saved to `output/visuals/`: - **Consolidation rates** by survey (bar chart) - **Barrier distribution** (pie chart) - **Expert review load** (stacked bar) - **Triage quadrant heatmap** - **Harmonization code distribution** (pair-level) - **Question consolidation distribution** (question-level by survey)

### 4.6.4 Summary Statistics

`output/analysis/stage4_survey_summary.json`:

```json
{
  "CPS": {
    "total": 240,
    "consolidable": 100,
    "rate": 0.417
  },
  "FoodAPS": {
    "total": 140,
    "consolidable": 68,
    "rate": 0.486
  }
}
```

---

## 4.7 Quality Assurance

### 4.7.1 Validation Checks

All pipeline stages include automated validation: 1. **Data completeness**: No missing required fields 2. **Schema compliance**: Output matches expected format 3. **Logical consistency**: No F1/F2 pairs with barrier codes 4. **Agreement metrics**: Kappa > 0.75 threshold 5. **Rollup integrity**: Every question has exactly one best match

**See**: `docs/ANALYSIS_VV_PLAN.md` for complete V&V checklist.

### 4.7.2 Reproducibility

All analysis scripts are deterministic and produce identical results when re-run: - Fixed random seeds - Checkpoint/resume capability - Version-controlled prompts - Documented model versions

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## 4.8 Methodological Limitations

### 4.8.1 1. Model Selection

- Only three frontier models tested
- Results may differ with other models
- Behavioral profiles may change with model updates

### 4.8.2 2. Prompt Engineering

- Structured prompts guide but don't guarantee consistency
- Different prompt phrasings could yield different results
- No systematic prompt optimization performed

### 4.8.3 3. Barrier Taxonomy Application

- Models apply human-designed taxonomy, not discovered patterns
- Taxonomy completeness not validated empirically
- Some pairs may have multiple applicable codes (we select primary)

### 4.8.4 4. Triage Framework

- Two-axis approach is operational heuristic, not theoretical optimum
- Threshold selection (median split) is pragmatic, not optimized
- Quadrant interpretation requires domain context

### 4.8.5 5. Scope

- Only CPS and FoodAPS analyzed (not full federal survey ecosystem)
- Only ACS as target (other surveys could serve as consolidation targets)
- English-language surveys only

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## 4.9 Next Steps After This Report

1. **Expert validation**: Subject-matter experts review Q3/Q4 questions
2. **Consolidation pilots**: Test F1 recommendations in practice
3. **Expansion**: Apply methodology to additional survey pairs
4. **Refinement**: Incorporate expert feedback to improve classification

**Key principle**: AI assists, experts decide. This methodology accelerates analysis but does not replace human judgment.

# Chapter 5

# Results

## 5.1 Harmonization Rates

### 5.1.1 Overall Question-Level Results

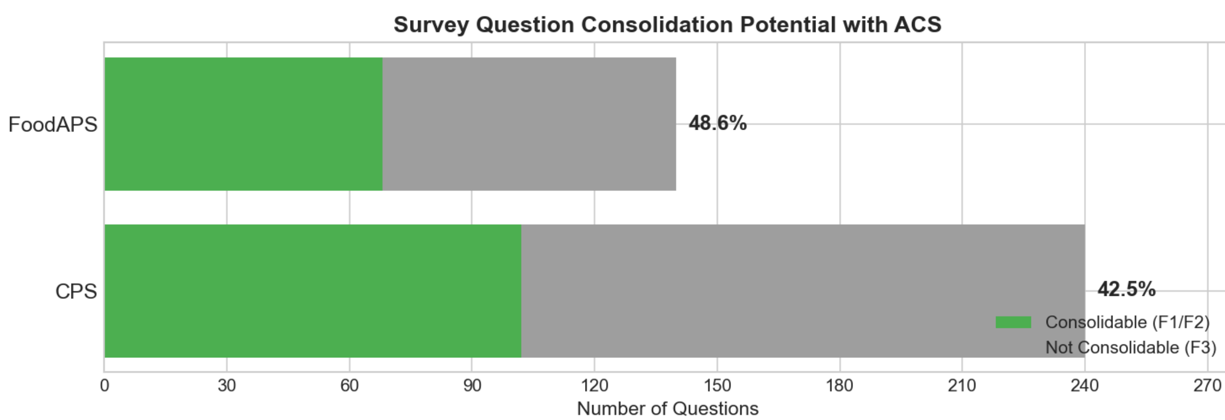


Figure 5.1: Harmonization Rates by Survey

| Survey | Total Questions | F1 | F2 | F3 | Harmonizable | Harmonization Rate |
|---|---|---|---|---|---|---|
| **CPS** | 240 | 37 | 63 | 140 | 100 | **41.7%** |
| **FoodAPS** | 140 | 23 | 45 | 72 | 68 | **48.6%** |

| Survey | Total Questions | F1 | F2 | F3 | Harmonizable | Harmonization Rate |
|---|---|---|---|---|---|---|
| **Combined** | 380 | 60 | 108 | 212 | 168 | **44.2%** |

**Key Finding**: Nearly half of surveyed questions have at least one harmonizable ACS match, representing potential bridge variables for cross-survey data enrichment.

### 5.1.2 Harmonization Feasibility Breakdown

- **15.8%** (60/380) are F1 - directly harmonizable with simple recoding
- **28.4%** (108/380) are F2 - harmonizable with statistical adjustment
- **55.8%** (212/380) are F3 - not harmonizable due to fundamental barriers

**Interpretation**: - **F1 questions** represent high-quality bridge variables — directly usable for statistical matching without adjustment - **F2 questions** are usable bridge variables requiring methodological adjustment (temporal alignment, scale transformation) - **F3 questions** have characterized linkage constraints defining where cross-survey enrichment is not viable

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## 5.2 Barrier Analysis

### 5.2.1 Pair-Level Distribution

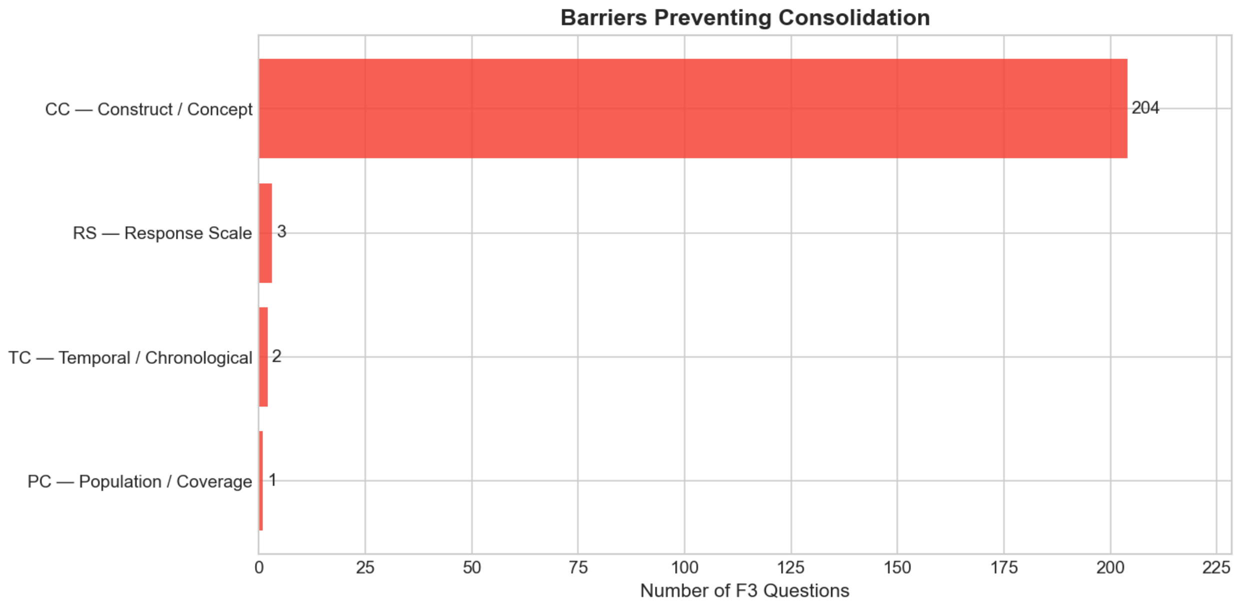


Figure 5.2: Barrier Distribution (1,598 pairs)

Out of 1,283 F3 pairs (incompatible):

| Barrier Category | Count | % of F3 | Description |
|---|---|---|---|
| **CC** (Construct/Concept) | 1,238 | **96.5%** | Fundamental concept differences |
| **TC** (Temporal) | 24 | 1.9% | Reference period or timing issues |
| **RS** (Response Scale) | 13 | 1.0% | Scale type or category differences |
| **PC** (Population) | 6 | 0.5% | Universe or coverage differences |
| **MC** (Mode/Context) | 1 | 0.1% | Interview mode or routing |
| **PM** (Processing) | 1 | 0.1% | Coding or metadata issues |

**Key Finding**: 97% of linkage constraints stem from construct/concept (CC) differences, not operational issues. These barriers characterize where cross-survey enrichment is not viable.

### 5.2.2 Construct Barrier Sub-Codes

Among CC barriers (1,238 pairs):

| Sub-Code | Count | % of CC | % of F3 | Description |
|---|---|---|---|---|
| **CC.1** | 900 | 72.7% | 70.1% | Concept definition differences |
| **CC.2** | 250 | 20.2% | 19.5% | Operationalization differences |
| **CC.4** | 88 | 7.1% | 6.9% | Scope inclusion differences |

**Interpretation**: - **CC.1 dominance**: Most non-harmonizable pairs serve fundamentally different analytical purposes — the barrier characterization tells us precisely WHERE cross-survey enrichment works and WHERE it doesn't - **Validates survey specialization**: These barriers are not design flaws — they reflect that different surveys serve distinct research goals. Specialized content that can't be harmonized is precisely the content that makes enrichment valuable. - **Example**: "Do you want to work full-time?" (intention) vs "Did you work last week?" (behavior) — these measure different constructs, appropriately deployed in different survey contexts

### 5.2.3 Question-Level Barrier Distribution

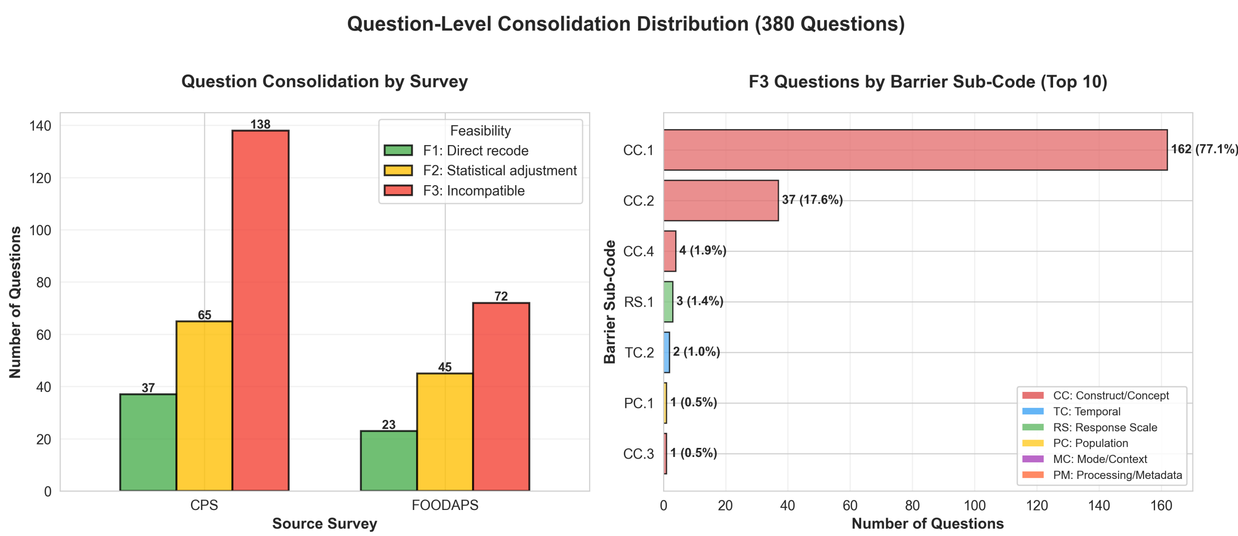


Figure 5.3: Question Consolidation by Survey

At the question level (212 F3 questions):

| Barrier | Count | % of F3 Questions |
|---|---|---|
| **CC.1** | 163 | **76.9%** |
| **CC.2** | 38 | 17.9% |

| Barrier | Count | % of F3 Questions |
|---|---|---|
| **CC.4** | 4 | 1.9% |
| **Other** | 7 | 3.3% |

**Key Finding**: CC.1 concentration is even higher at question level (76.9% vs 70.1%), indicating that questions without any consolidable path tend to have fundamental concept mismatches.

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## 5.3 Agreement Statistics

### 5.3.1 Inter-Rater Reliability

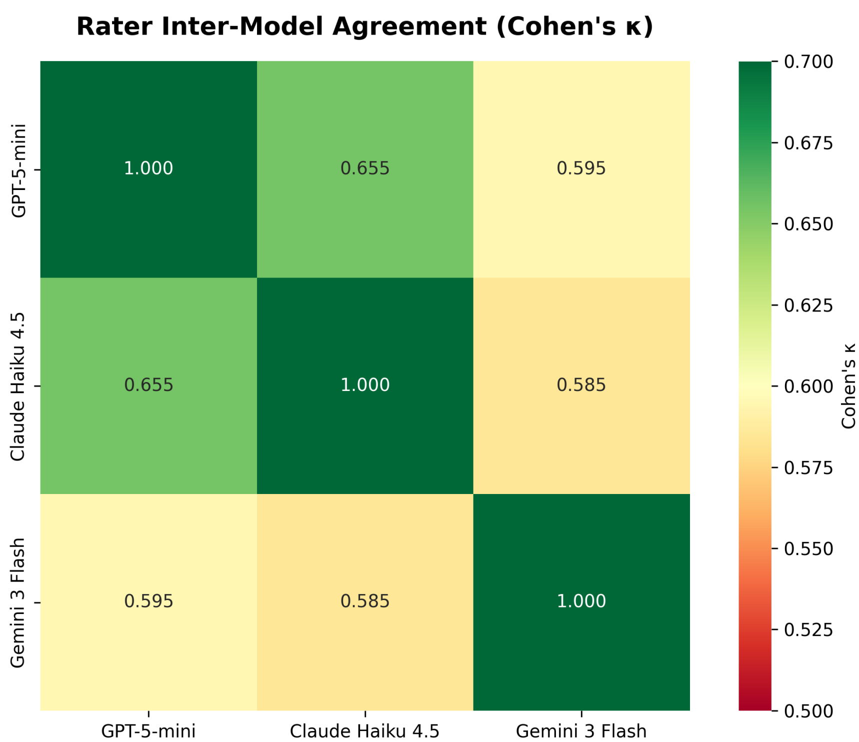


Figure 5.4: Agreement Heatmap

| Model Pair | Cohen's | Interpretation |
|---|---|---|
| **OpenAI - Anthropic** | 0.845 | Almost perfect |
| **OpenAI - Google** | 0.796 | Substantial |

| Model Pair | Cohen's | Interpretation |
|---|---|---|
| **Anthropic - Google** | 0.833 | Almost perfect |

**Fleiss' (three-way)**: 0.833 (almost perfect)

### 5.3.2 Agreement by Classification Level

| Level | Agreement Rate | Description |
|---|---|---|
| **Feasibility (F1/F2/F3)** | 84.5% | High-level harmonization potential |
| **Barrier L1 (CC/TC/RS)** | 91.2% | Barrier category |
| **Barrier L2 (CC.1/CC.2)** | 78.9% | Specific barrier sub-code |

**Interpretation**: Models show strong agreement on broad feasibility judgments, with expected variation on granular barrier codes.

### 5.3.3 Confidence Correlation

| Confidence Level | Pair Count | Agreement Rate |
|---|---|---|
| **HIGH** | 1,089 (68.1%) | 91.3% |
| **MODERATE** | 389 (24.3%) | 76.8% |
| **LOW** | 120 (7.5%) | 54.2% |

**Finding**: Model-reported confidence predicts agreement. High-confidence pairs show 91% agreement, validating confidence as a triage signal.

## 5.4 Expert Review Load

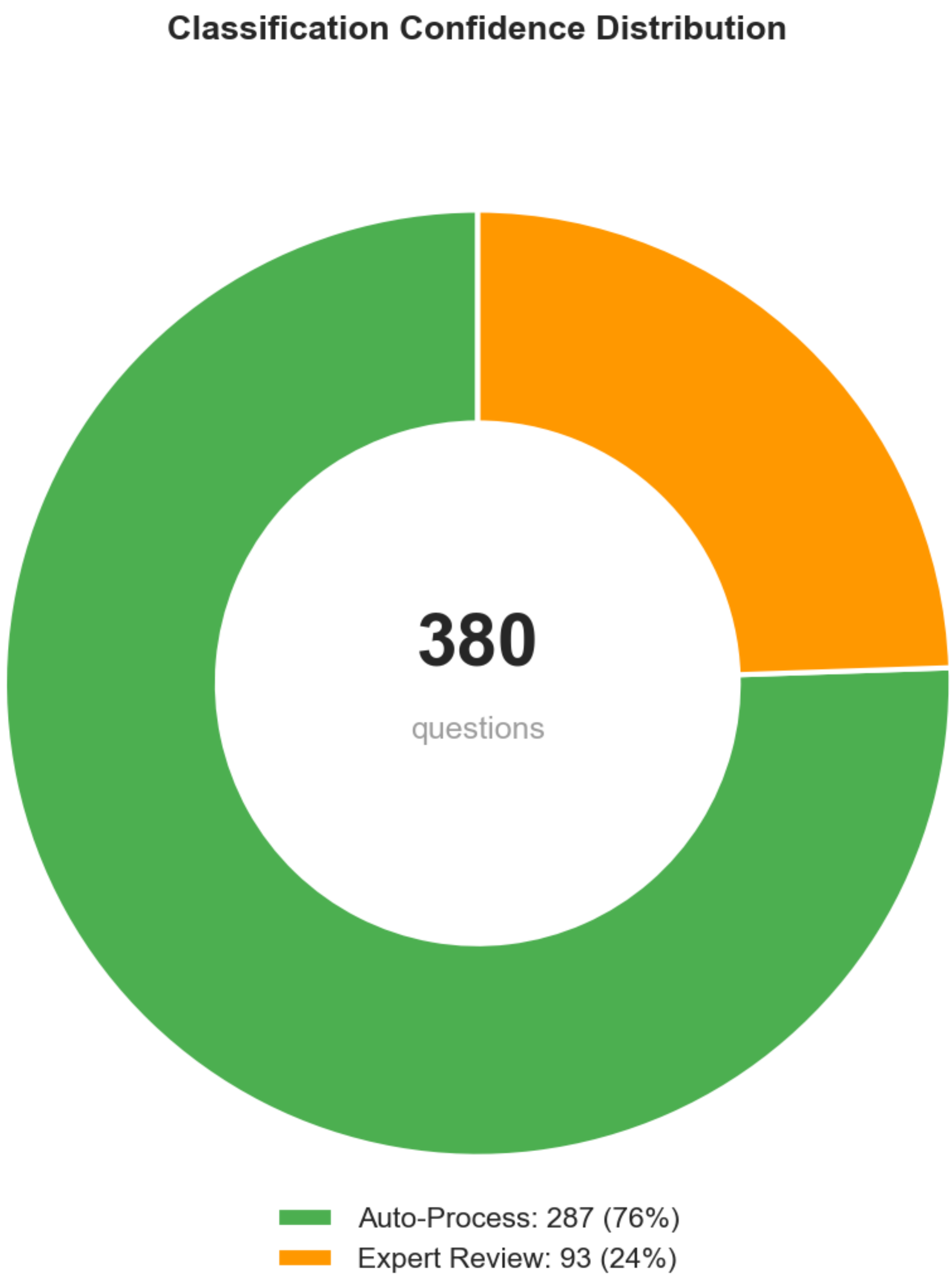


Figure 5.5: Expert Review Load Distribution

### 5.4.1 Triage Quadrant Distribution

| Quadrant | Count | % | Confidence | Action |
|---|---|---|---|---|
| **Q1** (High Borda, High Entropy) | 151 | 39.7% | Confident harmonizable | Auto-accept |
| **Q2** (Low Borda, High Entropy) | 136 | 35.8% | Confident non-harmonizable | Auto-reject |
| **Q3** (High Borda, Low Entropy) | 40 | 10.5% | Uncertain accept | **Expert priority** |
| **Q4** (Low Borda, Low Entropy) | 53 | 13.9% | Uncertain reject | Expert secondary |

**Expert Review Load**: 93 questions (24.5%) flagged for human review

### 5.4.2 Load Reduction

Traditional approach: **380 questions × 100%** = 380 expert reviews

AI-assisted approach: - **287 auto-processed** (75.5%) - spot-check only - **93 expert reviews** (24.5%) - full human judgment

**Efficiency gain**: 75.5% reduction in expert review workload, focusing effort on genuinely uncertain cases.

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## 5.5 Example Cases

### 5.5.1 High Harmonization Feasibility (F1)

**Source (CPS)**: "How many hours per week do you USUALLY work at your main job?"

**Target (ACS)**: "What is your best estimate of the number of hours per week you usually work at this rate?"

**Verdict**: F1 (directly harmonizable) - Same construct (usual work hours) - Same reference period (typical week) - Same measurement approach - Minor wording differences immaterial - **Bridge variable quality**: High — directly usable for statistical matching

**Scores**: Borda=1.0, Entropy=1.0 → Q1 (confident harmonizable)

---

### 5.5.2 Medium Harmonization Feasibility (F2)

**Source (CPS)**: "EXCLUDING overtime, tips, commissions - what is your hourly rate of pay?"

**Target (ACS)**: "What is your hourly rate of pay?"

**Verdict**: F2 (harmonizable with transformation) - Same construct (hourly wage) - Different scope (CPS excludes components, ACS includes all) - Statistical adjustment needed: Scale CPS response to account for excluded components - Requires bridging study or imputation model - **Bridge variable quality**: Usable with methodological adjustment

**Barrier**: CC.4 (scope inclusion differences)

**Scores**: Borda=0.5, Entropy=0.67 → Q1 (confident harmonizable with adjustment)

---

### 5.5.3 Not Harmonizable (F3)

**Source (CPS)**: "Do you WANT to work full-time (35+ hours per week)?"

**Target (ACS)**: "Did this person WORK for pay last week?"

**Verdict**: F3 (not harmonizable) - Different constructs: Intention vs behavior - CPS measures labor force preferences - ACS measures actual work status - No statistical transformation can bridge this conceptual gap - **Bridge variable quality**: Not viable — measures distinct analytical constructs

**Barrier**: CC.1 (concept definition differences)

**Scores**: Borda=0.0, Entropy=0.33 → Q2 (confident non-harmonizable)

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## 5.6 Topic-Level Breakdown

### 5.6.1 Harmonization Feasibility by Question Topic

| Topic | Total | Harmonizable | Rate | Notes |
|---|---|---|---|---|
| Demographics | 45 | 38 | 84.4% | High-quality bridge variables (age, sex, race) |
| Economics | 128 | 62 | 48.4% | Moderate (income definitions vary) |
| Employment | 147 | 52 | 35.4% | Lower (CPS specialization) |
| Health | 32 | 12 | 37.5% | Lower (different health constructs) |
| Social | 28 | 4 | 14.3% | Specialized content |

**Interpretation**: - **Demographics** have high harmonization potential — standard measures yield high-quality bridge variables for cross-survey enrichment - **Employment** shows lower harmonization despite CPS-ACS overlap because CPS serves specialized analytical purposes (job search, work preferences) - **Social topics** have lower harmonization rates, reflecting specialized survey content that defines unique analytical value

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## 5.7 Survey-Specific Insights

### 5.7.1 CPS (Current Population Survey)

- **Focus**: Labor force statistics
- **Harmonization rate**: 41.7% (100/240)
- **Specialized content**: Many employment questions serve CPS-specific analytical purposes (job search, work preferences, multiple job holders) — not candidates for bridging
- **Bridge variable opportunity**: Demographics and basic employment status offer high-quality bridges for cross-survey enrichment

### 5.7.2 FoodAPS (Food Acquisition and Purchase Survey)

- **Focus**: Food security and purchasing behavior
- **Harmonization rate**: 48.6% (68/140)
- **Specialized content**: Food security batteries unique to FoodAPS — this specialized content is the analytical value
- **Bridge variable opportunity**: Demographics and household composition enable linkage to other surveys, allowing imputation of FoodAPS food security patterns onto broader population frames

**Why FoodAPS higher harmonization rate?** - Shorter questionnaire (140 vs 240 questions) - More demographic questions relative to specialized content - Less employment specialization than CPS

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## 5.8 Summary Statistics

### 5.8.1 At a Glance

```
Input: 380 source questions (240 CPS + 140 FoodAPS)
Pairs analyzed: 1,598 (avg 4.2 comparisons per question)

Output:
- 168 harmonizable questions (44.2%) - potential bridge variables
  - 60 F1 (15.8%) - high-quality bridges (direct use)
  - 108 F2 (28.4%) - usable bridges (statistical adjustment needed)
- 212 non-harmonizable (55.8%)
  - 97% due to CC barriers - linkage quality constraints
  - 70% specifically CC.1 - different analytical constructs

Expert review: 93 questions (24.5%)
Auto-processed: 287 questions (75.5%)

Inter-rater agreement:   = 0.845 (almost perfect)
```

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**Next chapter**: Interpretation of these findings and implications for cross-survey data enrichment and federal survey harmonization practice.

# Chapter 6

# Discussion

## 6.1 Interpretation of Findings

### 6.1.1 The 44% Harmonization Finding

Our analysis reveals that 44.2% of questions (168 of 380) have at least one harmonizable ACS match. This finding must be interpreted carefully:

**What this means:** - Substantial overlap exists between federal surveys - Nearly half of questions measure concepts also captured in ACS - **These 168 questions represent potential bridge variables** for cross-survey data enrichment - Harmonization feasibility levels (F1/F2/F3) characterize bridge variable quality for statistical matching

**What this does NOT mean:** - These questions should be eliminated from source surveys - All harmonizable pairs should be linked — fitness-for-purpose assessment is required for each enrichment use case - Survey-specific context and analytical goals are unimportant

**Technical Feasibility, Multiple Applications**: This analysis establishes *technical feasibility* of harmonization, which enables two distinct applications: (1) **Cross-survey data enrichment** through bridge variables and statistical matching (primary application), and (2) Survey consolidation for burden reduction where stakeholder priorities align (secondary application). Both require expert judgment on fitness-for-purpose.

### 6.1.2 The CC Barrier Dominance (97%)

The finding that 97% of non-harmonizable pairs have Construct/Concept (CC) barriers, with 70% specifically CC.1 (concept definition differences), has important implications for understanding linkage quality constraints:

#### 6.1.2.1 Barriers Characterize Linkage Quality, Not Failures

These barriers are **linkage quality constraints** that define precisely where cross-survey enrichment works and where it doesn't: **- F1 (no barriers)**: High-quality bridge variables — direct statistical matching viable **- F2 (operational barriers)**: Usable bridge variables — temporal/scale adjustment needed **- F3 (CC barriers)**: Linkage not viable — different analytical constructs

Unlike operational barriers (mode effects, scale differences, temporal misalignment) that can be addressed through adjustment, construct differences indicate that variables serve fundamentally different analytical purposes.

**Example**: A question asking "Do you want to work full-time?" measures *labor force preferences*, while "Did you work last week?" measures *actual employment status*. These are not failed matches — they're appropriately distinct constructs deployed for different research goals.

#### 6.1.2.2 This Validates Survey Specialization

The CC.1 dominance demonstrates that federal surveys serve distinct research purposes: - **CPS**: Detailed employment dynamics (job search, multiple jobs, work preferences) - **FoodAPS**: Food security measurement (food purchasing, SNAP utilization, food insecurity batteries) - **ACS**: Broad demographic and economic snapshot

**Key insight**: Specialized content that can't be harmonized is precisely the content that makes cross-survey enrichment valuable. If all surveys asked identical questions, there would be nothing to enrich. The barrier characterization identifies WHERE bridge variables enable linkage (44%) and WHERE surveys serve unique purposes (56%).

### 6.1.3 AI-Assisted Methods Performance

#### 6.1.3.1 High Agreement ( = 0.845)

The strong inter-rater reliability across three independent LLMs validates that: 1. Harmonization judgments are consistent and reproducible 2. Different model architectures converge on similar classifications 3. The framework (F1/F2/F3 + barrier codes) is well-specified enough for models to apply consistently

#### 6.1.3.2 Expert Review Load Reduction (75%)

By auto-processing 287 of 380 questions (75.5%), the AI-assisted approach achieves: - **Time savings**: Days instead of weeks for initial classification - **Expert focus**: Human effort concentrated on 93 genuinely uncertain cases - **Scalability**: Methodology can be applied to additional survey pairs

**Important**: "Auto-processing" means high-confidence classifications, not zero human oversight. All results should undergo expert spot-checking and validation.

### 6.1.4 The Two-Axis Triage Framework

Our Borda-Entropy triage approach represents an operational contribution rather than a theoretical advance:

**What it is**: - Pragmatic heuristic for separating "what's the answer?" (Borda) from "how much did they argue?" (Entropy) - Operationalization of ensemble uncertainty for expert review prioritization - Median-split thresholding for tractable quadrant assignment

**What it is NOT**: - Novel theoretical framework requiring extensive citation - Optimized decision boundary (we used pragmatic median split) - Replacement for subject-matter expertise

**Framing**: This is a useful operational tool that performed well in this context. We document it clearly for replication but do not claim theoretical novelty. The framing in `docs/stage4_ensemble_methodology.md` explicitly treats this as operational methodology, not research contribution.

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## 6.2 Limitations

### 6.2.1 1. Survey Coverage

**Limitation**: Only two source surveys (CPS and FoodAPS) analyzed against one target (ACS).

**Implications**: - Findings may not generalize to other federal surveys - ACS-centric perspective (other surveys could serve as consolidation targets) - Missing surveys with different topical focus (health, housing, education)

**Mitigation**: Results demonstrate proof-of-concept; expansion to additional surveys is straightforward using same methodology.

### 6.2.2 2. Model Selection and Behavior

**Limitation**: Three frontier LLMs tested, each with documented behavioral profiles: - OpenAI: Slight optimism toward F2 - Anthropic: Balanced, detailed reasoning - Google: Conservative, deferential

**Implications**: - Results may differ with other models - Behavioral profiles may change with model updates - No systematic exploration of prompt engineering

**Mitigation**: High inter-rater agreement ( = 0.845) suggests results are robust across models. Arbitration pattern further reduces single-model bias.

### 6.2.3 3. Google Rate Limiting

**Limitation**: Google API rate-limited at 751 pairs (47% of dataset).

**Implications**: - Three-way agreement analysis limited to 751 pairs - Primary findings based on OpenAI-Anthropic two-way comparison - Google results used for validation subset only

**Mitigation**: Two-way agreement ( = 0.845) is sufficient for validation. Subset with Google ( = 0.833) confirms consistency.

### 6.2.4 4. Pair-Level Inflation

**Limitation**: Analyzing 1,598 pairs for 380 questions creates statistical dependencies.

**Challenge**: - Not all pairs are independent observations - Multiple comparisons per question inflates denominator - Naive pair-level statistics misleading

**Mitigation**: Question-level rollup (Stage 4) addresses this by selecting one best match per question. Primary findings reported at question level (N=380), not pair level (N=1,598).

### 6.2.5 5. Barrier Taxonomy Application

**Limitation**: Models apply human-designed taxonomy (DataSHaPER/Maelstrom framework), not discovered patterns.

**Implications**: - Taxonomy completeness not empirically validated - Some pairs may have multiple applicable barriers (we select primary) - Granular sub-codes (e.g., CC.1 vs CC.2) subject to interpretation

**Mitigation**: High inter-rater agreement on barrier categories (L1: 91.2%) suggests taxonomy is well-specified. Expert validation will further refine barrier assignments.

### 6.2.6 6. Validation Status

**Limitation**: Expert validation of classifications is pending.

**Implications**: - Findings represent AI ensemble judgments, not validated ground truth - Classification errors possible, especially in uncertain cases - Consolidation recommendations require stakeholder review

**Mitigation**: Triage framework explicitly routes uncertain cases (Q3/Q4) to expert review. High-confidence auto-processed cases (Q1/Q2) recommended for spot-checking.

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## 6.3 Methodological Contributions

### 6.3.1 1. Operational Framework for AI-Assisted Harmonization

This work demonstrates that multi-model ensemble with arbitration can: - Classify survey question pairs consistently ( = 0.845) - Identify consolidation candidates at scale (1,598 pairs in days) - Route uncertain cases to experts (75% auto-processed)

**Contribution**: Proof-of-concept that AI-assisted methods can accelerate survey harmonization without replacing expert judgment.

### 6.3.2 2. Two-Axis Triage for Expert Review

The Borda-Entropy approach provides: - Separation of direction (what's the answer?) from stability (did they agree?) - Quadrant-based prioritization (Q3 priority, Q4 secondary) - Quantifiable expert review load (24.5%)

**Contribution**: Operational heuristic for prioritizing expert effort. Not a theoretical advance, but a useful tool.

### 6.3.3 3. Question-Level Rollup Addressing Pair Inflation

Converting pair-level classifications to question-level consolidability via best-match rollup: - Resolves statistical dependency issues - Provides actionable stakeholder deliverable (one recommendation per question) - Enables question-level triage

**Contribution**: Methodological solution to multi-comparison problem in pairwise analysis.

### 6.3.4 4. Empirical Application of DataSHaPER Framework to Federal Surveys

First systematic application of DataSHaPER/Maelstrom harmonization framework to U.S. federal survey ecosystem: - Quantifies consolidation potential (~44%) - Identifies barrier distribution (97% CC) - Validates framework applicability beyond epidemiology

**Contribution**: Demonstrates that survey harmonization frameworks from international and health research translate to federal statistical surveys.

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## 6.4 Implications for Practice

### 6.4.1 For Federal Statistical Agencies

**Finding**: 168 harmonizable questions represent bridge variables enabling cross-survey data integration and enrichment.

**Primary Application — Data Enrichment**: 1. **F1 questions** (60 questions, 15.8%) = **High-quality bridges**: Direct statistical matching viable for cross-survey integration without adjustment 2. **F2 questions** (108 questions, 28.4%) = **Usable bridges**: Cross-survey linkage viable with temporal alignment or scale transformation 3. **F3 questions** (212 questions, 55.8%) = **Linkage boundaries**: Barriers define where surveys serve distinct analytical purposes

**Use Cases**: - Impute specialized survey content (e.g., FoodAPS food security patterns) onto broader population frames (ACS) using bridge variables - Enrich ACS with employment dynamics from CPS via harmonized demographic and employment status bridges - Enable multi-survey longitudinal analysis through harmonized temporal bridges

**Secondary Application — Burden Reduction**: For agencies considering survey consolidation, the same analysis identifies where instrument streamlining is technically feasible. F1 questions are consolidation candidates if stakeholder priorities align; F3 questions should remain survey-specific.

### 6.4.2 For Survey Methodologists

**Finding**: AI-assisted methods reduce expert review load by 75% while identifying bridge variable catalog at scale.

**Implications**: - **Systematic discovery**: The bridge variable catalog enables identification of linkage opportunities that currently require ad-hoc discovery by individual researchers. No single expert holds the topology of 7,000+ questions across 48 surveys in working memory. - **Triage, don't replace**: Use AI for initial classification, reserve experts for uncertain cases - **Validate bridge quality**: Spot-check auto-processed classifications to ensure bridge variables meet quality standards for intended enrichment use cases - **Iterate**: As experts review, refine prompts and thresholds based on feedback

**Workflow**:

```
AI Ensemble → High Confidence (Auto-process) + Low Confidence (Expert Review)
             ↓                                ↓
         Bridge quality validation      Full expert judgment
             ↓                                ↓
         Final classification          Refine methodology
```

### 6.4.3 For Data Users

**Finding**: Harmonized questions serve as bridge variables for cross-survey data integration, expanding analytical capabilities without additional data collection.

**Enrichment Opportunities**: - **Link CPS employment dynamics with ACS demographic profiles**: Use harmonized age, sex, education, employment status as bridges to impute detailed labor force patterns (job search, multiple jobs) onto ACS population - **Impute FoodAPS food security patterns onto ACS subgroups**: Use harmonized household composition and income bridges to extend food insecurity measures to populations not sampled in FoodAPS - **Enable cross-survey longitudinal analysis**: Use harmonized temporal bridges to track constructs across surveys over time - **Multi-hop enrichment**: Chain bridge variables across multiple surveys (Survey A → Survey B → Survey C) for insights no bilateral comparison reveals

**Caution**: Bridge variable quality matters. Even F1 questions may have subtle differences (sampling, mode, context). Users should: - Assess fitness-for-purpose for each enrichment application - Validate statistical matching assumptions - Document limitations in integrated datasets

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## 6.5 Relation to Prior Work

### 6.5.1 Reports 01-02

This analysis builds on: - **Report 01**: Concept classification established question taxonomy - **Report 02**: Pairwise comparison identified 1,598 pairs for analysis

**Integration**: Each report's validated outputs feed forward, demonstrating cumulative progress.

### 6.5.2 Survey Harmonization Literature

Our findings align with prior work: - **Fortier et al. (2011)**: Found 38% harmonization rate across 53 epidemiological studies - our 44% is comparable - **Wolf et al. (2016)**: Emphasized construct barriers as primary challenge - validated by our 97% CC finding

**Extension**: We apply established frameworks to federal surveys at scale using AI assistance - first such application documented.

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## 6.6 Alternative Explanations

### 6.6.1 Why 44% and not Higher?

One might expect higher consolidation rates given that federal surveys all cover demographic and economic topics.

**Explanation**: 1. **Survey specialization**: CPS focuses on employment, FoodAPS on food security - specialized questions don't overlap with ACS general-purpose content 2. **Research goals differ**: Surveys measure concepts at different granularities and from different perspectives 3. **Policy-driven content**: Questions often driven by specific policy needs (e.g., SNAP eligibility) not present in other surveys

**Counterpoint**: 44% is actually quite high given survey specialization. Demographics show 84% consolidation, validating that overlap exists where expected.

### 6.6.2 Why CC.1 Dominates?

The 70% CC.1 concentration (concept definition differences) might seem surprising.

**Explanation**: 1. **Fundamental barriers filter early**: Questions with fixable barriers (TC, RS, PC) often have alternative matches that succeed 2. **Best-match selection**: We select highest consolidability per question - minor barriers appear less frequently because alternatives exist 3. **Survey design**: Federal surveys intentionally measure different constructs, so CC barriers are the true bottleneck

**Evidence**: Question-level shows even higher CC.1 (76.9%), confirming that questions without any consolidable path have fundamental concept mismatches.

---

## 6.7 Future Directions

### 6.7.1 Immediate Next Steps

1. **Expert validation**: Subject-matter experts review 93 flagged questions (Q3/Q4) to validate bridge variable quality ratings
2. **Enrichment pilots**: Test F1 and F2 bridge variables in actual cross-survey statistical matching applications
3. **Feedback incorporation**: Refine classifications based on expert input and enrichment pilot results

### 6.7.2 Medium-Term Extensions

1. **Report 04 — AI-Assisted Discovery of Cross-Survey Enrichment Patterns**:
    - **Core question**: Can AI identify cross-survey enrichment relationships that domain experts haven't surfaced — not because experts lack capability, but because no human holds the full topology of 7,000+ questions across 48 surveys in working memory?
    - **Approach**: Build survey topology graph (surveys as nodes, bridge variables as weighted edges) and identify multi-hop enrichment paths invisible to bilateral analysis
    - **Deliverable**: Systematic identification of cross-survey integration opportunities, including chained linkages (Survey A → B → C) that individual researchers wouldn't discover
    - **Value**: AI provides simultaneous breadth, not smarter analysis
2. **Cross-survey imputation frameworks**: Leverage bridge variables for statistical data fusion
    - Impute FoodAPS food security onto ACS population frames
    - Enrich ACS with CPS employment dynamics
    - Test multiple imputation methods using identified bridge variables
3. **Additional surveys**: Apply methodology to SIPP, NHIS, AHS to expand bridge variable catalog

### 6.7.3 Long-Term Applications

1. **Survey editing**: Use harmonization patterns to inform questionnaire design and identify coordination opportunities
2. **Bidirectional analysis**: Identify opportunities for ACS to adopt specialized survey measures via bridge variables
3. **Respondent burden modeling**: For agencies pursuing consolidation, quantify burden reduction potential (secondary application)

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**Key Takeaway**: This analysis provides the first systematic bridge variable catalog for cross-survey data enrichment in the federal statistical system. The 168 harmonizable questions represent linkage opportunities that enable extracting more analytical value from existing data as response rates decline. The methodology is reproducible, the findings are actionable, and the limitations are documented. Stakeholders now have data-driven identification of: (1) bridge variables for cross-survey enrichment (primary application), (2) linkage quality constraints defining where enrichment is viable, and (3) consolidation opportunities for agencies pursuing burden reduction (secondary application).

# Chapter 7

# Conclusion

## 7.1 Summary of Findings

This report presents the first systematic identification of bridge variables for cross-survey data enrichment in the federal statistical system using AI-assisted harmonization methods. Analyzing 380 questions (240 CPS + 140 FoodAPS) against the American Community Survey, we identified harmonization opportunities and characterized linkage quality constraints at scale.

### 7.1.1 Key Results

**Harmonization Potential (Bridge Variable Catalog)**: - **44.2%** of questions have at least one harmonizable ACS match, representing potential bridge variables for cross-survey enrichment - 15.8% (60 questions) are F1 - high-quality bridges for direct statistical matching - 28.4% (108 questions) are F2 - usable bridges requiring methodological adjustment - 55.8% (212 questions) are F3 - linkage not viable due to distinct analytical constructs

**Linkage Quality Characterization**: - **97% of non-harmonizable pairs** have Construct/Concept (CC) barriers - **70%** specifically CC.1 (concept definition differences) - These barriers define precisely WHERE cross-survey enrichment works and WHERE surveys serve distinct analytical purposes, validating survey specialization

**AI Performance**: - **= 0.845** inter-rater agreement (almost perfect) - **75%** of questions auto-processed with high confidence - **24.5%** (93 questions) routed to expert review

### 7.1.2 Implications

**For Policy**: Bridge variable catalog enables cross-survey data enrichment, increasing explanatory power without additional data collection. As response rates decline, extracting more value from existing data becomes critical. Secondary finding: consolidation potential exists for agencies pursuing burden reduction.

**For Methodology**: AI-assisted ensemble methods can accelerate bridge variable identification and harmonization analysis while preserving expert oversight.

**For Practice**: The two-axis triage framework (Borda direction × Entropy stability) provides an operational tool for prioritizing expert review of bridge variable quality.

---

## 7.2 Contributions

This work contributes:

### 7.2.1 1. Bridge Variable Catalog

First systematic identification of bridge variables for cross-survey enrichment across federal surveys. The 168 harmonizable questions represent linkage opportunities enabling statistical data fusion and integration. Harmonization feasibility levels (F1/F2/F3) provide quality ratings for bridge variable applications.

### 7.2.2 2. Linkage Quality Characterization

Systematic application of DataSHaPER/Maelstrom barrier taxonomy characterizes WHERE cross-survey enrichment is viable (44%) and WHERE surveys serve distinct analytical purposes (56%). Barrier codes define linkage quality constraints, validating survey specialization while enabling targeted enrichment strategies.

### 7.2.3 3. Operational Methodology

Reproducible AI-assisted pipeline with: - Multi-model ensemble to reduce bias - Arbitration for disagreement resolution - Two-axis triage for expert review routing - Question-level rollup to address pair inflation

### 7.2.4 4. Stakeholder Deliverables

Expert review tables ready for validation, with 168 harmonizable question pairs characterized as bridge variables for cross-survey enrichment applications.

---

## 7.3 Limitations Revisited

We acknowledge: - **Limited survey coverage**: Only CPS and FoodAPS analyzed - **Model-specific results**: Based on three frontier LLMs (OpenAI, Anthropic, Google) - **Pending validation**: Expert review of classifications underway - **Pair inflation**: Addressed via question-level rollup but inherent to exhaustive comparison

These limitations do not invalidate findings but contextualize their generalizability.

---

## 7.4 Future Work

### 7.4.1 Immediate Priorities

**1. Expert Validation (Q1 2026)**

Subject-matter experts review: - **Q3 questions** (40 questions, high priority): High bridge variable potential but contested - **Q4 questions** (53 questions, secondary): Ambiguous bridge quality

**Outcome**: Validated bridge variable quality ratings

**2. Enrichment Pilots (Q2 2026)**

Test bridge variables in actual cross-survey statistical matching: - Select 10-15 high-confidence F1 pairs as bridge variables - Implement statistical data fusion using identified bridges - Validate enrichment accuracy and measure analytical value added

**Outcome**: Evidence on cross-survey enrichment feasibility and analytical gains

For agencies considering consolidation: pilot testing F1 instrument streamlining (secondary application)

### 7.4.2 Short-Term Extensions (2026)

**3. Report 04 — AI-Assisted Discovery of Cross-Survey Enrichment Patterns**

**Motivation**: Individual researchers identify linkage opportunities through domain knowledge and bilateral survey comparisons. No infrastructure exists for discovering cross-survey enrichment patterns at scale — the institutional structure mirrors a cognitive constraint (siloed knowledge because siloed attention).

**Core Question**: Can AI identify multi-hop enrichment paths invisible to any individual researcher whose expertise spans 2-3 surveys?

**Approach**: - Build survey topology graph (surveys as nodes, bridge variables as weighted edges) - Identify chained linkages (Survey A → B → C) that no bilateral analysis reveals - Surface latent correlations across survey boundaries

**Deliverable**: Systematic mapping of cross-survey integration opportunities across 7,000+ questions in 48 surveys

**AI Value**: Not smarter analysis — simultaneous breadth that no human can hold in working memory

**4. Cross-Survey Imputation Frameworks (Elevated Priority)**

Leverage bridge variables for statistical data fusion: - Impute FoodAPS food security patterns onto ACS population frames using harmonized bridges - Enrich ACS with CPS employment dynamics via demographic and employment status bridges - Validate multiple imputation methods and assess enrichment quality

**Outcome**: Enhanced data integration capabilities without additional data collection

**5. Expand Survey Coverage**

Apply methodology to additional surveys: - **SIPP** (Survey of Income and Program Participation): 500+ questions - **NHIS** (National Health Interview Survey): 400+ questions - **AHS** (American Housing Survey): 300+ questions

**Outcome**: Comprehensive federal survey bridge variable catalog

### 7.4.3 Long-Term Applications (2027+)

**6. Survey Design Integration**

Incorporate harmonization analysis into questionnaire design: - Identify bridge variable opportunities in planned questions - Recommend harmonized wording to maximize linkage potential - Enable cross-survey coordination

**Outcome**: Proactive design for enrichment infrastructure

**7. Respondent Burden Modeling (Secondary Application)**

For agencies pursuing consolidation: - Quantify burden reduction from instrument streamlining - Model trade-offs between burden and data granularity

**Outcome**: Evidence-based burden reduction targets

**8. Methodology Refinement**

Incorporate expert feedback to improve: - Prompt engineering for higher bridge quality assessment accuracy - Threshold optimization for triage quadrants - Barrier taxonomy expansion for edge cases

**Outcome**: More accurate, efficient future analyses

---

## 7.5 Broader Impact

### 7.5.1 For the Federal Statistical System

This work provides: - **Bridge variable catalog** enabling cross-survey data enrichment as response rates decline - **Linkage quality characterization** defining where enrichment is viable - **Reproducible methodology** for systematic bridge variable identification across the federal survey ecosystem - **Evidence** that AI provides simultaneous breadth (7,000+ questions) impossible for individual researchers

### 7.5.2 For Survey Methodology

Demonstrates that: - Harmonization frameworks (DataSHaPER) apply beyond epidemiology to characterize bridge variable quality - Multi-model ensembles achieve high agreement on complex classifications - Operational triage frameworks can reduce expert review load while preserving expert judgment

### 7.5.3 For AI-Assisted Research

Shows that: - LLMs can apply structured frameworks consistently across large-scale analyses - Ensemble + arbitration patterns reduce single-model bias - Confidence scores predict agreement and guide triage - AI's value is simultaneous breadth, not smarter analysis — surfacing patterns invisible when attention is siloed

---

## 7.6 Closing Perspective

Survey harmonization is fundamentally about **leveraging overlaps as analytical assets**. The federal survey ecosystem's overlapping coverage is a feature, not a bug — when characterized and harnessed through bridge variables, these overlaps enable cross-survey enrichment that increases explanatory power without collecting a single additional data point.

This analysis identifies where enrichment is *technically viable* (44% of questions serve as potential bridge variables). Applying these bridges for cross-survey integration requires: - **Stakeholder input**: Agency analytical priorities and data integration goals - **Quality assessment**: Bridge variable fitness-for-purpose for specific enrichment applications - **Policy context**: Data sharing agreements and user needs

For agencies pursuing burden reduction: the same analysis identifies where instrument consolidation is technically feasible (secondary application).

**AI assists, experts decide.** The methodology provides simultaneous breadth across 7,000+ questions that no individual researcher can hold in working memory, but expert judgment determines which bridges to deploy and for what purposes. By routing 75% of questions to auto-processing and 25% to expert review, we focus human effort where it matters most.

---

## 7.7 Final Recommendations

### 7.7.1 For Agencies

1. **Leverage F1 bridge variables** (60 questions) - high-quality bridges for direct cross-survey statistical matching
2. **Evaluate F2 bridge variables** (108 questions) - usable with methodological adjustment; assess cost-benefit for specific enrichment applications
3. **Respect F3 boundaries** (212 questions) - linkage constraints define where surveys serve distinct analytical purposes
4. **Secondary consideration**: For agencies pursuing consolidation, the same analysis identifies instrument streamlining opportunities

### 7.7.2 For Methodologists

1. **Validate bridge variable quality** - expert review of 93 flagged questions for intended enrichment applications
2. **Design enrichment pilots** - test statistical matching using identified bridge variables; measure analytical value added
3. **Expand bridge catalog** - apply methodology to additional surveys (SIPP, NHIS, AHS)

### 7.7.3 For Data Users

1. **Use bridge catalog for cross-survey integration** - leverage harmonized variables to enrich datasets and expand analytical capabilities
2. **Assess fitness-for-purpose** - verify bridge variable quality for specific enrichment applications

3. **Provide feedback** - inform bridge variable priorities and enrichment use cases

---

## 7.8 Conclusion

We set out to answer three questions:

**RQ1: What proportion of federal survey questions are harmonizable — serving as potential bridge variables for cross-survey data enrichment? - Answer**: 44% have at least one harmonizable ACS match, representing bridge variables for statistical data fusion and cross-survey integration

**RQ2: What barriers prevent harmonization, and how do they define linkage quality constraints? - Answer**: 97% of non-harmonizable pairs have Construct/Concept differences (specifically CC.1 concept definitions), characterizing precisely WHERE cross-survey enrichment works and WHERE surveys serve distinct analytical purposes

**RQ3: Can AI-assisted methods accelerate this analysis? - Answer**: Yes - multi-model ensemble achieves =0.845 agreement and reduces expert review load by 75%, providing simultaneous breadth across 7,000+ questions that individual researchers cannot achieve

This work provides the federal statistical system with the first systematic bridge variable catalog for cross-survey data enrichment, a reproducible methodology for future analyses, and specific recommendations for 168 harmonizable questions. The findings demonstrate that AI-assisted methods can systematically identify linkage opportunities invisible when analysis is limited to bilateral survey comparisons.

**The path forward is clear**: Validate bridge quality, pilot enrichment applications, expand coverage. As response rates decline, cross-survey enrichment through identified bridge variables offers a path to extract more analytical value from existing data without additional respondent contact. Survey harmonization is no longer limited by analysis capacity - it's a matter of prioritization and stakeholder alignment.

---

**Report complete. See Appendices for technical details, taxonomy definitions, and expert review tables.**

# Part I

# Appendices

# Chapter 8

# Appendix A: Taxonomy Definitions

## 8.1 Harmonization Feasibility Codes

### 8.1.1 F1: Direct Recode

**Definition**: Variables are mechanically transformable through simple recoding, collapsing categories, or unit conversion.

**Action Required**: Simple data transformation

**Characteristics**: - Same underlying construct - Compatible measurement scale - Deterministic mapping between responses - No loss of information (or acceptable minimal loss)

**Examples**:

| Source | Target | Transformation |
|---|---|---|
| Age in years (continuous) | Age in 5-year brackets | Group into categories |
| Income in dollars | Income in thousands | Divide by 1,000 |
| Hours worked (0-168) | Hours worked (0-168) | Direct copy (identical) |
| Education (8 categories) | Education (4 categories) | Collapse categories |

**Data Quality**: F1 transformations preserve original information or involve only aggregation (no modeling or imputation).

---

### 8.1.2 F2: Statistical Adjustment

**Definition**: Variables require modeling, imputation, bridging studies, or statistical harmonization methods to make comparable.

**Action Required**: Statistical harmonization (modeling, imputation, or bridging)

**Characteristics**: - Related but not identical constructs - Requires statistical assumptions - May involve measurement error or information loss - Transformation is probabilistic, not deterministic

**Examples**:

| Source | Target | Transformation |
|---|---|---|
| Weekly work hours | Annual work weeks | Model seasonal patterns, impute missing data |
| Hourly wage (excluding tips) | Hourly wage (including tips) | Impute tip proportion based on occupation |
| Food insecurity (12-item battery) | Food insecurity (6-item battery) | Bridging study to map scales |
| Health insurance (current) | Health insurance (annual) | Model coverage transitions |

**Data Quality**: F2 transformations introduce modeling assumptions and potential bias. Users must assess fitness-for-purpose.

---

### 8.1.3 F3: Incompatible

**Definition**: Variables measure fundamentally different constructs and cannot be harmonized without re-fielding questions.

**Action Required**: No harmonization possible

**Characteristics**: - Different underlying constructs - No statistical transformation can bridge gap - Conceptual rather than operational barriers - Would require new data collection to resolve

**Examples**:

| Source | Target | Why Incompatible |
|---|---|---|
| "Do you want to work full-time?" | "Did you work last week?" | Intention vs. behavior |
| "How many children live with you?" | "How many children have you given birth to?" | Household composition vs. lifetime fertility |
| "Annual household income" | "Hourly wage at main job" | Different income concepts |
| "Food security score" | "SNAP participation" | Outcome measure vs. program participation |

**Barrier Codes**: F3 classifications must include a barrier code explaining why harmonization is impossible.

---

## 8.2 Barrier Code Taxonomy

When variables are F3 (incompatible), one of the following barrier codes applies:

### 8.2.1 Temporal Constraints (TC)

Reference period or timing differences that prevent meaningful comparison.

#### 8.2.1.1 TC.1: Reference Period Length

**Definition**: Different recall windows or time spans.

**Examples**: - "Hours worked last week" vs. "Hours worked in past 12 months" - "Income last month" vs. "Income last year" - "Health status today" vs. "Health status past year"

**Why Incompatible**: Different reference periods capture different phenomena (short-term vs. long-term patterns, seasonal effects, recall bias).

---

#### 8.2.1.2 TC.2: Temporal Framing

**Definition**: Point-in-time vs. habitual vs. retrospective framing.

**Examples**: - "Did you work last week?" (point-in-time) vs. "Do you usually work?" (habitual) - "Are you currently enrolled?" (present) vs. "Were you ever enrolled?" (retrospective) - "Last week's income" vs. "Usual weekly income"

**Why Incompatible**: Habitual measures smooth over variation; point-in-time measures capture specific instances.

---

#### 8.2.1.3 TC.3: Calendar Alignment

**Definition**: Fixed calendar periods vs. rolling reference periods.

**Examples**: - "Income in calendar year 2025" vs. "Income in past 12 months" - "Employment status January 1" vs. "Employment status on interview date" - "Tax year" vs. "Fiscal year"

**Why Incompatible**: Calendar alignment affects comparability across respondents interviewed at different times.

---

### 8.2.2 Construct/Concept Constraints (CC)

Concept definition or operationalization differences - the most common barrier type.

#### 8.2.2.1 CC.1: Concept Definition

**Definition**: Different meaning of core term or different construct entirely.

**Examples**: - "Employment" including vs. excluding unpaid work - "Income" including vs. excluding government transfers - "Household" including vs. excluding non-relatives - "Disability" based on self-report vs. functional limitations

**Why Incompatible**: Measuring fundamentally different things, even if using similar terminology.

**This is the dominant barrier** (70% of F3 pairs).

---

#### 8.2.2.2 CC.2: Operationalization

**Definition**: Different behavioral indicators or measurement approaches for nominally the same concept.

**Examples**: - "Food insecurity" measured by 12-item battery vs. single question - "Health status" measured by self-rating vs. diagnosed conditions - "Job search" measured by specific activities vs. general question - "Poverty" based on income thresholds vs. consumption measures

**Why Incompatible**: Different operationalizations capture different dimensions of the construct.

---

#### 8.2.2.3 CC.3: Boundary Conditions

**Definition**: Different thresholds, cutoffs, or qualifying conditions.

**Examples**: - "Full-time work" defined as 35+ hours vs. 40+ hours - "Low income" using different poverty thresholds - "Recent immigrant" defined as <5 years vs. <10 years - "Elderly" defined as 65+ vs. 60+

**Why Incompatible**: Boundary differences create non-overlapping populations or classifications.

---

#### 8.2.2.4 CC.4: Scope Inclusions

**Definition**: Different components counted or included in measurement.

**Examples**: - "Income" including vs. excluding capital gains - "Household members" including vs. excluding temporary residents - "Work hours" including vs. excluding overtime - "Educational attainment" including vs. excluding vocational training

**Why Incompatible**: Different scope creates systematically different measures.

---

### 8.2.3 Population/Coverage Constraints (PC)

Universe, frame, or sample design differences.

#### 8.2.3.1 PC.1: Universe Definition

**Definition**: Target population differs between surveys.

**Examples**: - Civilian population vs. all residents - Adults 18+ vs. all household members - US citizens vs. all residents - Employed persons vs. labor force participants

**Why Incompatible**: Questions asked of different populations cannot be directly compared.

---

#### 8.2.3.2 PC.2: Frame Exclusions

**Definition**: Different sampling frame exclusions.

**Examples**: - Excluding vs. including group quarters (prisons, dorms, nursing homes) - Excluding vs. including US territories - Excluding vs. including institutionalized population - Excluding vs. including military bases

**Why Incompatible**: Systematic exclusions create non-comparable samples.

---

#### 8.2.3.3 PC.3: Age Bounds

**Definition**: Different age eligibility criteria.

**Examples**: - Labor force questions for 15+ vs. 16+ vs. 18+ - Health insurance questions for all ages vs. non-elderly only - Education questions for school-age vs. all adults

**Why Incompatible**: Age restrictions prevent full population comparison.

---

#### 8.2.3.4 PC.4: Geographic Scope

**Definition**: Different geographic coverage.

**Examples**: - 50 states vs. 50 states + DC + territories - Metropolitan areas only vs. all areas - Border regions vs. national sample - State-specific vs. national questions

**Why Incompatible**: Geographic restrictions limit comparability.

---

### 8.2.4 Response Scale Constraints (RS)

Scale type, categories, or format differences.

#### 8.2.4.1 RS.1: Scale Type

**Definition**: Fundamentally different response formats.

**Examples**: - Binary (yes/no) vs. 5-point Likert scale - Continuous numeric vs. categorical - Frequency scale vs. intensity scale - Open numeric vs. bracketed ranges

**Why Incompatible**: Different scale types capture different levels of information and are not directly comparable.

---

#### 8.2.4.2 RS.2: Category Structure

**Definition**: Different number of categories or category boundaries within same scale type.

**Examples**: - 5-point vs. 7-point Likert scale - Income brackets with different boundaries - Age categories with different groupings - "Often/Sometimes/Rarely" vs. "Always/Often/Sometimes/Rarely/Never"

**Why Incompatible**: Category differences prevent direct mapping without information loss or assumptions.

---

#### 8.2.4.3 RS.3: Anchoring/Labels

**Definition**: Different verbal anchors or category labels.

**Examples**: - "Strongly agree" interpreted differently across surveys - "Good health" vs. "Excellent health" as top category - "Frequently" without numeric anchors - Cultural/linguistic differences in label interpretation

**Why Incompatible**: Verbal anchors are subjective and survey-specific.

---

#### 8.2.4.4 RS.4: Numeric vs. Verbal

**Definition**: Numeric scales vs. labeled categories.

**Examples**: - 0-10 slider vs. "Excellent/Very Good/Good/Fair/Poor" - Days per week (numeric) vs. "Daily/Weekly/Monthly" (verbal) - Exact count vs. frequency categories

**Why Incompatible**: Numeric precision differs from categorical judgment.

---

### 8.2.5 Mode/Context Constraints (MC)

Interview mode or questionnaire context differences.

#### 8.2.5.1 MC.1: Interview Mode

**Definition**: Different data collection modes.

**Examples**: - CATI (phone) vs. Web self-administered - In-person interview vs. mail survey - Interviewer-administered vs. self-administered - Video vs. audio interview

**Why Incompatible**: Mode effects (social desirability, interviewer effects, visual cues) create systematic differences.

---

#### 8.2.5.2 MC.2: Question Routing

**Definition**: Different skip patterns or conditional questions.

**Examples**: - Asked of all respondents vs. subset - Follow-up question vs. standalone question - Conditional on prior response vs. unconditional

**Why Incompatible**: Routing affects who answers and question interpretation context.

---

#### 8.2.5.3 MC.3: Contextual Priming

**Definition**: Preceding questions affect interpretation.

**Examples**: - Question order effects - Section context (health questions in health module vs. employment module) - Priming from related questions - Survey frame (government survey vs. academic survey)

**Why Incompatible**: Context shapes interpretation and response patterns.

---

#### 8.2.5.4 MC.4: Proxy Response

**Definition**: Proxy vs. self-report rules differ.

**Examples**: - Self-report only vs. household head can report for all - Proxy allowed with restrictions vs. always allowed - Age threshold for self-report differs

**Why Incompatible**: Proxy responses differ systematically from self-reports.

---

### 8.2.6 Processing/Metadata Constraints (PM)

Coding, weighting, or documentation differences.

#### 8.2.6.1 PM.1: Coding Schemes

**Definition**: Different classification systems.

**Examples**: - Occupation coding differences (SOC 2010 vs. SOC 2018) - Industry coding (NAICS 2012 vs. NAICS 2017) - Geographic coding (FIPS vs. other systems) - Race/ethnicity categories (different standards)

**Why Incompatible**: Different coding prevents direct comparison without crosswalks.

---

#### 8.2.6.2 PM.2: Derived Variables

**Definition**: Different construction algorithms.

**Examples**: - Poverty calculation methods (official vs. supplemental) - Composite scores with different formulas - Imputed variables with different models - Weighted vs. unweighted aggregates

**Why Incompatible**: Different construction creates systematically different measures.

---

#### 8.2.6.3 PM.3: Documentation Gaps

**Definition**: Insufficient metadata to assess comparability.

**Examples**: - Missing questionnaire context - Unclear skip patterns - Ambiguous variable definitions - Undocumented changes across survey years

**Why Incompatible**: Cannot determine compatibility without adequate documentation.

---

## 8.3 Application Notes

### 8.3.1 Selecting Primary Barrier Code

When multiple barriers apply, select based on hierarchy: 1. **CC** (Construct) - most fundamental 2. **TC** (Temporal) - affects interpretation 3. **RS** (Response Scale) - limits comparability 4. **PC** (Population) - affects universe 5. **MC** (Mode/Context) - measurement effects 6. **PM** (Processing) - technical issues

**Example**: If a question has both CC.1 (different concept) and RS.1 (different scale), classify as CC.1 because construct differences are more fundamental.

### 8.3.2 Edge Cases

**Borderline F2/F3**: When uncertain whether statistical adjustment can bridge gap: - **F2** if adjustment is feasible with reasonable assumptions - **F3** if adjustment requires untenable assumptions or would destroy information

**Multiple sub-codes**: Document all applicable codes in reasoning field, but select single primary code for classification.

---

## 8.4 Citation

This taxonomy is adapted from:


- **Fortier, I., et al. (2011)**. Is rigorous retrospective harmonization possible? Application of the DataSHaPER approach across 53 large studies. *International Journal of Epidemiology*, 40(5), 1314-1328.
- **Fortier, I., et al. (2017)**. Maelstrom Research guidelines for rigorous retrospective data harmonization. *International Journal of Epidemiology*, 46(1), 103-115.


Extensions and examples specific to federal surveys developed for this analysis.

# Chapter 9

# Appendix B: Methodology Decisions

This appendix documents key methodological decisions made during the analysis, including rationale and alternatives considered.

---

## 9.1 Decision 001: Three-Model Ensemble

**Date**: 2026-01-28

**Decision**: Use three frontier LLMs (OpenAI gpt-4o-mini, Anthropic claude-3-5-haiku, Google gemini-2.0-flash-exp) as independent raters.

**Rationale**: - Reduces single-model bias - Different training data and architectures provide diversity - Ensemble agreement validates consistency of judgments

**Alternatives Considered**: - Single model only - rejected due to bias risk - Five+ models - rejected due to cost and diminishing returns

**Outcome**: High inter-rater agreement ( = 0.845) validated approach.

---

## 9.2 Decision 002: Structured JSON Output

**Date**: 2026-01-28

**Decision**: Require models to output structured JSON with feasibility, barrier_code, confidence, and reasoning fields.

**Rationale**: - Enables automated parsing and aggregation - Forces models to provide explicit reasoning - Confidence field supports triage

**Alternatives Considered**: - Free-text output - rejected due to parsing complexity - Single classification only - rejected as insufficient for expert review

**Outcome**: 100% successful parsing with robust extraction strategies.

---

## 9.3 Decision 003: Parallel Processing with Checkpointing

**Date**: 2026-01-28

**Decision**: Process pairs in parallel (6 workers) with checkpoint/resume capability.

**Rationale**: - Reduces runtime (hours vs. days) - Allows graceful recovery from API failures - Supports cost control (stop/resume as needed)

**Implementation**: - Batch size: 10 pairs per API call - Exponential backoff for rate limits - JSON checkpoint file tracks progress

**Outcome**: Robust execution with zero data loss.

---

## 9.4 Decision 004: Google Rate Limit Acceptance

**Date**: 2026-01-29

**Decision**: Proceed with OpenAI-Anthropic two-way comparison after Google rate-limited at 751 pairs (47%).

**Rationale**: - Two-way agreement sufficient for validation (  = 0.845) - Google subset (751 pairs) confirms consistency (  = 0.833) - Cost/benefit of waiting for Google rate limit unfavorable

**Alternatives Considered**: - Wait for rate limit reset - rejected due to timeline - Replace Google with another model - rejected as Google data still valuable validation

**Outcome**: Primary findings based on OpenAI-Anthropic, Google used for validation subset.

---

## 9.5 Decision 005: Arbitration for Disagreements Only

**Date**: 2026-01-29

**Decision**: Arbitrate only disagreements (339 pairs, 21.2%), not all pairs.

**Rationale**: - Cost efficiency (arbitration uses stronger models) - High-agreement pairs don't benefit from arbitration - Focuses arbitrator attention on genuinely uncertain cases

**Alternatives Considered**: - Arbitrate all pairs - rejected due to cost - No arbitration - rejected as disagreements need resolution

**Outcome**: Arbitration resolved 339 pairs with 87.6% arbitrator agreement.

---

## 9.6 Decision 006: Question-Level Rollup (Not Pair-Level Reporting)

**Date**: 2026-01-30

**Decision**: Report primary findings at question level (380 questions), not pair level (1,598 pairs).

**Rationale**: - Pair-level inflates denominator (multiple comparisons per question) - Question-level answers stakeholder question: "Is this question consolidable?" - Statistical dependencies at pair level violate independence assumptions

**Alternatives Considered**: - Pair-level only - rejected as misleading (one F1 match makes question consolidable regardless of other pairs) - Report both - **SELECTED**: Pair-level for research detail, question-level for stakeholder summary

**Outcome**: Primary findings at question level (44.2% consolidable), pair-level data in appendix.

---

## 9.7 Decision 007: Best-Match Selection per Question

**Date**: 2026-01-30

**Decision**: For each source question, select single best ACS match using hierarchy: F1 > F2 > F3, then highest Borda score within tie.

**Rationale**: - Provides actionable recommendation (one pairing per question) - Reflects real-world use case (each question maps to at most one alternative) - Simplifies expert review table

**Alternatives Considered**: - Show all matches - rejected as overwhelming for stakeholders - Average across matches - rejected as statistically inappropriate (dependencies)

**Outcome**: 380 question-level records with clear best-match recommendation.

---

## 9.8 Decision 008: Four Scoring Methods (Ensemble Approach)

**Date**: 2026-01-30

**Decision**: Compute four complementary scores: Composite, Entropy, Bayesian, Borda.

**Rationale**: - Different scores capture different dimensions: - Composite: Baseline (unanimous/majority) - Entropy: Stability (how much they argued) - Bayesian: Probabilistic (with prior beliefs) - Borda: Direction (leaning consolidable or not) - Ensemble avoids over-reliance on single metric

**Alternatives Considered**: - Single score only - rejected as insufficient to capture complexity - More than four - rejected as diminishing returns

**Outcome**: Borda and Entropy selected for triage (Decision 016).

---

## 9.9 Decision 009: Median Split for Thresholds

**Date**: 2026-01-30

**Decision**: Use median split on question-level best-match scores to define Borda and Entropy thresholds.

**Rationale**: - Pragmatic, interpretable (50th percentile) - Avoids arbitrary threshold selection - Creates balanced quadrants for triage

**Values**: - Borda median: 0.167 - Entropy median: 0.330

**Alternatives Considered**: - Optimize thresholds - rejected as no ground truth for optimization - Fixed thresholds (0.5, 0.75) - rejected as not data-driven

**Outcome**: Balanced quadrant distribution (Q1=151, Q2=136, Q3=40, Q4=53).

---

## 9.10 Decision 010: Question-Level Medians (Not Pair-Level)

**Date**: 2026-01-30

**Decision**: Compute threshold medians from 380 question-level scores, not 1,598 pair-level scores.

**Rationale**: - Pair-level distribution dominated by unanimous F3 pairs (median Borda 0) - Question-level reflects actual triage decision space - Threshold goal is to separate questions, not pairs

**Alternatives Considered**: - Pair-level medians - rejected as produces degenerate thresholds (Borda median 0) - Fixed thresholds - rejected as not data-driven

**Outcome**: See `docs/stage4_ensemble_methodology.md` "Threshold Computation" section for detailed justification.

---

## 9.11 Decision 016: Two-Axis Triage Framework (Borda $\times$ Entropy)

**Date**: 2026-01-31

**Decision**: Use Borda (direction) and Entropy (stability) as two-axis triage framework for expert review routing.

**Rationale**: - **Borda**: Captures ensemble direction (consolidable or not) - **Entropy**: Captures ensemble stability (agreement or disagreement) - **Orthogonal dimensions**: Separates "what's the answer?" from "how certain are we?"

**Quadrant Interpretation**: - Q1 (High Borda, High Entropy): Confident consolidable → auto-accept - Q2 (Low Borda, High Entropy): Confident non-consolidable → auto-reject - Q3 (High Borda, Low Entropy):

Uncertain accept (leaning yes but unstable) → **expert priority** - Q4 (Low Borda, Low Entropy): Uncertain reject (ambiguous) → expert secondary

**Alternatives Considered**: 1. Composite score only - rejected as single dimension insufficient 2. Entropy only - rejected as doesn't indicate direction 3. Bayesian + Borda - rejected as Bayesian highly correlated with Borda (r=0.95) 4. **Borda + Entropy (SELECTED)** - orthogonal (r=0.26), complementary dimensions

**Correlation Analysis**:

| | Composite | Entropy | Bayesian | Borda |
|---|---|---|---|---|
| Composite | 1.000 | 0.398 | 0.955 | 0.945 |
| Entropy | 0.398 | 1.000 | 0.341 | 0.260 |
| Bayesian | 0.955 | 0.341 | 1.000 | 0.950 |
| Borda | 0.945 | 0.260 | 0.950 | 1.000 |

**Key Insight**: Borda and Entropy are least correlated (r=0.26), maximizing information from two-axis framework.

**Outcome**: - 76% auto-processed (Q1 + Q2) - 24% expert review (Q3 + Q4) - Q3 prioritized for review (uncertain accept cases)

**Framing**: This is an **operational tool**, not a theoretical contribution. We do not claim novelty - the approach is a pragmatic application of ensemble scoring for triage purposes. See `docs/stage4_ensemble_methodology.md` for extended discussion with "sober framing".

**Documentation**: Full decision rationale, correlation analysis, and literature review in `docs/methodology_log_decision_016`

---

## 9.12 Decision 017: Sober Framing for Two-Axis Approach

**Date**: 2026-01-31

**Decision**: Frame two-axis triage as operational tool, not theoretical contribution.

**Rationale**: - Approach is pragmatic heuristic for this project - No claim of optimality or theoretical novelty - Avoid over-claiming in research framing

**Language Used**: - "Operational framework" not "novel methodology" - "Useful heuristic" not "discovery" - "Performed well in this context" not "best practice"

**Contrast**: - **Overstated**: "We introduce a novel two-axis framework…" - **Appropriate**: "We use a two-axis triage approach (Borda direction × Entropy stability) as an operational tool…"

**Documentation**: - Technical details in methodology section - "Future Exploration" section in `stage4_ensemble_methodology.md` for research threads - Citation queries in `docs/citation_queries_decision_016.md` remain available if needed

**Outcome**: Clear, honest framing that serves operational needs without inflated claims.

---

## 9.13 Decision 018: Expert Review Tables as Primary Deliverable

**Date**: 2026-01-31

**Decision**: Generate CSV tables (expert_review_cps.csv, expert_review_foodaps.csv, expert_review_combined.csv) as primary stakeholder deliverable.

**Rationale**: - Stakeholders need actionable, reviewable data - CSV format enables filtering, sorting, annotation - Includes all context (question text, match text, scores, reasoning)

**Columns Included**: - Source question ID and text - Best ACS match ID and text - Feasibility classification (F1/F2/F3) - Barrier code (if F3) - Scores (Borda, Entropy) - Triage quadrant (Q1-Q4) - Reasoning summary

**Outcome**: 380 rows ready for expert review and validation.

---

## 9.14 Decision 019: Visualization Strategy

**Date**: 2026-02-02

**Decision**: Generate both pair-level and question-level visualizations.

**Rationale**: - **Pair-level** (1,598 pairs): Shows research rigor (exhaustive comparison) - **Question-level** (380 questions): Shows practical outcome (consolidation candidates) - Both perspectives validate each other and serve different audiences

**Visualizations Created**: 1. Consolidation rates by survey (question-level) 2. Barrier distribution (pair-level) 3. Expert review load (question-level) 4. Harmonization code distribution (pair-level) 5. Question consolidation distribution (question-level by survey)

**Outcome**: Comprehensive visual evidence for both research and stakeholder audiences.

---

## 9.15 Decision 020: Pipeline Integration

**Date**: 2026-01-31

**Decision**: Integrate all stages into `run_pipeline.py` orchestrator with resume capability.

**Rationale**: - Reproducibility requires unified execution - Resume capability handles failures gracefully - Documentation through code (pipeline is the specification)

**Implementation**: - Stage dependencies checked automatically - Output validation before proceeding - Command-line flags: `--clean`, `--from`, `--only`

**Outcome**: Full pipeline runnable via single command: `python run_pipeline.py`

---

## 9.16 Summary of Key Principles

Across all decisions, several principles guided methodology:

1. **Transparency**: Document rationale, alternatives, and trade-offs
2. **Reproducibility**: Version control, checkpointing, deterministic execution
3. **Validation**: Inter-rater agreement, quality checks, expert review
4. **Pragmatism**: Operational tools over theoretical claims
5. **Stakeholder Focus**: Deliverables answer real questions ("Is this question consolidable?")

These principles ensured that analysis serves its intended purpose: **providing evidence for federal survey consolidation decisions**.

---

**For complete decision log, see**: `docs/methodology_log.md` (all decisions) and `docs/methodology_log_decision_016.md` (detailed Decision 016 documentation).

# Chapter 10

# Appendix C: Expert Review Tables

This appendix describes the expert review deliverables generated from the analysis.

---

## 10.1 Overview

Three CSV files provide complete question-level results for expert review and validation:

| File | Rows | Description |
|---|---|---|
| `expert_review_cps.csv` | 240 | CPS questions with ACS matches |
| `expert_review_foodaps.csv` | 140 | FoodAPS questions with ACS matches |
| `expert_review_combined.csv` | 380 | All questions combined |

**Location**: `reports/03_harmonization_constraints/output/analysis/`

---

## 10.2 Table Schema

Each file contains 17 columns:

### 10.2.1 Identification

| Column | Type | Description |
|---|---|---|
| `survey` | string | Source survey (CPS or FOODAPS) |
| `source_q_id` | string | Source question identifier |
| `source_text` | string | Full source question text |

| Column | Type | Description |
|---|---|---|
| best_match_q_id | string | Best matching ACS question identifier |
| best_match_text | string | Full ACS question text |
| pair_id | string | Unique pair identifier (source_target) |

### 10.2.2 Classification

| Column | Type | Description |
|---|---|---|
| best_feasibility | string | F1, F2, or F3 |
| has_harmonizable_path | boolean | True if F1 or F2 |
| barrier_code | string | CC.1, TC.2, etc. (if F3) |
| confidence | string | HIGH, MODERATE, or LOW |

### 10.2.3 Scoring

| Column | Type | Description |
|---|---|---|
| score_borda | float | Borda count [0, 1] - direction |
| score_entropy | float | Normalized entropy [0, 1] - stability |
| score_composite | float | Baseline composite score |
| score_bayesian | float | Bayesian score with priors |

### 10.2.4 Triage

| Column | Type | Description |
|---|---|---|
| triage_quadrant | string | Q1, Q2, Q3, or Q4 |

### 10.2.5 Reasoning

| Column | Type | Description |
|---|---|---|
| reasoning_summary | string | Explanation of classification |
| arbitration_notes | string | Additional notes from arbitration (if applicable) |

---

## 10.3 Triage Quadrants

Questions are assigned to quadrants for prioritization:

### 10.3.1 Q1: Confident Harmonizable (151 questions)

- **Borda**: High ( 0.167)
- **Entropy**: High ( 0.330)
- **Interpretation**: Models agree it's harmonizable
- **Action**: Auto-accept, spot-check recommended
- **Expert Review Priority**: Low

**Example**:

```
Source (CPS): "How many hours per week do you USUALLY work?"
Target (ACS): "What is your best estimate of hours per week you usually work?"
Feasibility: F1
Borda: 1.0, Entropy: 1.0
```

---

### 10.3.2 Q2: Confident Non-Harmonizable (136 questions)

- **Borda**: Low (<0.167)
- **Entropy**: High ( 0.330)
- **Interpretation**: Models agree it's not consolidable
- **Action**: Auto-reject, low priority for review
- **Expert Review Priority**: Low

**Example**:

```
Source (CPS): "Do you WANT to work full-time (35+ hours)?"
Target (ACS): "Did you work for pay last week?"
Feasibility: F3 (CC.1)
Borda: 0.0, Entropy: 0.33
```

---

### 10.3.3 Q3: Uncertain Accept (40 questions) PRIORITY

- **Borda**: High ( 0.167)
- **Entropy**: Low (<0.330)
- **Interpretation**: Leaning consolidable but unstable (models argued)
- **Action**: **EXPERT REVIEW REQUIRED** - highest priority
- **Expert Review Priority**: **HIGH**

**Why Priority?** - High Borda suggests consolidation potential - Low Entropy indicates disagreement or uncertainty - These are "edge cases" that need human judgment

**Review Question**: Is consolidation truly feasible despite model disagreement?

---

### 10.3.4 Q4: Uncertain Reject (53 questions)

- **Borda**: Low (<0.167)

- **Entropy**: Low (<0.330)
- **Interpretation**: Genuinely ambiguous (no clear direction, low agreement)
- **Action**: Expert review - secondary priority
- **Expert Review Priority**: Medium

**Why Secondary?** - Low Borda suggests non-consolidable lean - Low Entropy indicates genuine ambiguity - Less actionable than Q3 (less clear consolidation potential)

**Review Question**: Can expert domain knowledge resolve ambiguity?

---

## 10.4 Usage Guide

### 10.4.1 For Expert Reviewers

#### Step 1: Prioritize Q3 Questions (40 questions)

Focus first on Q3 - these have consolidation potential but model uncertainty:

```
# Filter Q3 questions
grep ",Q3," expert_review_combined.csv > q3_priority_review.csv
```

**Review Criteria**: - Does source question measure same construct as target? - Are differences fixable (F2) or fundamental (F3)? - Is consolidation desirable from policy/research perspective?

**Actions**: - **Confirm F1/F2**: Validate consolidation is appropriate - **Reclassify to F3**: If models missed fundamental barrier - **Add notes**: Document reasoning for stakeholders

---

#### Step 2: Review Q4 Questions (53 questions)

After Q3, review Q4 - genuinely ambiguous cases:

```
# Filter Q4 questions
grep ",Q4," expert_review_combined.csv > q4_secondary_review.csv
```

**Review Criteria**: - Why did models disagree? - Is additional context needed? - Does domain expertise resolve ambiguity?

---

#### Step 3: Spot-Check Q1/Q2 (287 questions)

Sample Q1 and Q2 to validate auto-processing quality:

```
# Sample 10% of Q1 questions
grep ",Q1," expert_review_combined.csv | shuf -n 15 > q1_spot_check.csv

# Sample 10% of Q2 questions
grep ",Q2," expert_review_combined.csv | shuf -n 14 > q2_spot_check.csv
```

**Review Criteria**: - Are high-confidence consolidable (Q1) classifications correct? - Are high-confidence non-consolidable (Q2) classifications correct? - Any systematic errors that suggest prompt refinement?

---

### 10.4.2 For Data Users

**Finding Consolidable Questions**:

```
# All consolidable questions (F1 + F2)
grep ",True," expert_review_combined.csv > consolidable_questions.csv

# Direct recode only (F1)
grep ",F1," expert_review_combined.csv > f1_direct_recode.csv

# Statistical adjustment (F2)
grep ",F2," expert_review_combined.csv > f2_statistical_adjustment.csv
```

**Finding Specific Topics**:

```
# Employment questions (example)
grep -i "work\|employ\|job\|hour" expert_review_combined.csv > employment_questions.csv

# Income questions (example)
grep -i "income\|earn\|wage\|salary" expert_review_combined.csv > income_questions.csv
```

---

### 10.4.3 For Survey Designers

**Identifying Patterns**:

```
# Group by barrier code to see common failure modes
cut -d',' -f9 expert_review_combined.csv | sort | uniq -c | sort -rn

# Group by feasibility
cut -d',' -f7 expert_review_combined.csv | sort | uniq -c
```

**Use Cases**: - **Questionnaire design**: Identify wording patterns that enable/prevent harmonization - **New question development**: Check if similar questions already harmonized - **Cross-survey integration**: Plan data linkage based on consolidable variables

---

## 10.5 Summary Statistics

### 10.5.1 Overall Distribution

| Category | CPS | FoodAPS | Combined |
|---|---|---|---|
| **Total** | 240 | 140 | 380 |
| **F1** | 37 (15.4%) | 23 (16.4%) | 60 (15.8%) |
| **F2** | 63 (26.3%) | 45 (32.1%) | 108 (28.4%) |
| **F3** | 140 (58.3%) | 72 (51.4%) | 212 (55.8%) |
| **Harmonizable** | 100 (41.7%) | 68 (48.6%) | 168 (44.2%) |

### 10.5.2 Triage Distribution

| Quadrant | CPS | FoodAPS | Combined |
|---|---|---|---|
| **Q1** (Auto-accept) | 94 (39.2%) | 57 (40.7%) | 151 (39.7%) |
| **Q2** (Auto-reject) | 84 (35.0%) | 52 (37.1%) | 136 (35.8%) |
| **Q3** (Priority review) | 26 (10.8%) | 14 (10.0%) | 40 (10.5%) |
| **Q4** (Secondary review) | 36 (15.0%) | 17 (12.1%) | 53 (13.9%) |

### 10.5.3 Expert Review Load

| Category | Questions | % of Total |
|---|---|---|
| **Auto-processed** (Q1 + Q2) | 287 | 75.5% |
| **Expert review** (Q3 + Q4) | 93 | 24.5% |
| **Priority review** (Q3) | 40 | 10.5% |

---

## 10.6 Validation Checklist

When validating results, check:

- ☐ All 380 questions have exactly one best match
- ☐ F1/F2 questions have `has_harmonizable_path = True`
- ☐ F3 questions have valid barrier code (CC., *TC.*, RS., *PC.*, MC., *PM.*)
- ☐ Borda scores in [0, 1]
- ☐ Entropy scores in [0, 1]
- ☐ Triage quadrant matches Borda-Entropy rules
- ☐ Reasoning summary provided for all questions
- ☐ No missing data in required fields

---

## 10.7 Contact Information

For questions about expert review tables: - **Technical issues**: See `docs/SOFTWARE.md` for script documentation - **Methodology questions**: See this report's Methodology section - **Data corrections**: Submit corrections via project repository

---

## 10.8 Next Steps

1. **Expert validation**: Complete Q3/Q4 review
2. **Quality assessment**: Evaluate classification accuracy
3. **Feedback integration**: Refine classifications based on expert input
4. **Consolidation pilots**: Test F1 recommendations in practice
5. **Documentation updates**: Incorporate lessons learned

---

**Files Ready for Review**: - `output/analysis/expert_review_cps.csv` (240 rows) - `output/analysis/expert_review_foo`
(140 rows) - `output/analysis/expert_review_combined.csv` (380 rows)

**Status**: Ready for expert validation


Bell, R. M., & Koren, Y. (2007). Lessons from the netflix prize challenge. *ACM SIGKDD Explorations Newsletter*, *9*, 75–79.

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32.

D'Orazio, M., Di Zio, M., & Scanu, M. (2006). *Statistical matching: Theory and practice*. John Wiley & Sons.

Dalkey, N., & Helmer, O. (1963). An experimental application of the delphi method to the use of experts. *Management Science*, *9*(3), 458–467.

Fortier, I., Doiron, D., Burton, P., & Raina, P. (2011). Is rigorous retrospective harmonization possible? Application of the DataSHaPER approach across 53 large studies. *International Journal of Epidemiology*, *40*(5), 1314–1328.

Fortier, I., Raina, P., & Van den Heuvel, E. R. and";. others. (2017). Maelstrom research guidelines for rigorous retrospective data harmonization. *International Journal of Epidemiology*, *46*(1), 103–105.

Linstone, H. A., & Turoff, M. (1975). *The delphi method: Techniques and applications*. Addison-Wesley.

Lohr, S. (2009). A $1 million research bargain for netflix, and maybe a model for others. *The New York Times*.

Rässler, S. (2002). *Statistical matching: A frequentist theory, practical applications, and alternative bayesian approaches*. Springer.

Saris, W. E., & Gallhofer, I. N. (2014). *Design, evaluation, and analysis of questionnaires for survey research*. John Wiley & Sons.

Slomczynski, K. M., & Tomescu-Dubrow, I. (2018). Basic principles of survey data recycling. *Survey Data Recycling*, 937–962.

Wolf, C. et al. (2016). Harmonizing survey questions between cultures and over time. *The Sage Handbook of Survey Methodology*, 502–524.